

Question #1 of 140

Question ID: 1473172

Which of the following statements regarding the P/E to growth (PEG) valuation approach is *least* accurate? The P/E to growth (PEG) valuation approach assumes that:

- A) there is a linear relationship between price to earnings (P/E) and growth. 
- B) stocks with higher PEGs are more attractive than stocks with lower PEGs. 
- C) there are no risk differences among stocks. 

Explanation

The PEG valuation approach implicitly assumes there is a linear relationship between price to earnings (P/E) and growth, even though there is not a "real world" linear relationship. The analyst must be cautious when using the PEG ratio for valuation or comparison purposes especially if the growth rate is very small or very large. If earnings or the growth rate is negative the PEG ratio is meaningless. The PEG ratio does not adjust for varying levels of risk among stocks and views stocks with lower PEG ratios to be more attractive than stocks with higher PEG ratios.

(Module 22.4, LOS 22.j)

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Question ID: 1473144

An increase in growth will cause a price-to-earnings (P/E) multiple to:

- A) there is insufficient information to tell. 
- B) decrease. 
- C) increase. 

Explanation

An increase in growth will decrease the denominator and increase the numerator in the trailing P/E expression, both of which should increase the P/E ratio:

$$P_0/E_0 = [(1 - b)(1 + g)] / (r - g)$$

Note that the topic review does not allow for any interactive relationship between retention and growth. Thus, no explicit consideration is given to how the growth increase was generated.

(Module 22.4, LOS 22.g)

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Question ID: 1473112

All other variables held constant, the justified price-to-book multiple will *decrease* with a decrease in:

- A) payout ratio. 

B) expected growth rate.



C) required rate of return.



Explanation

All other variables held constant, a decrease in expected growth rate will result in a decrease in the justified price-to-book multiple.

(Module 22.2, LOS 22.g)

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Question ID: 1473158

An analyst has gathered the following fundamental data:

	Firm A	Firm B	Firm C	Firm D
Payout Ratio	75%			
Required Rate of Return	12%	12%	12%	12%
Return on Equity (ROE)	20%	15%	30%	14%
Price/Book Value (PBV) Ratio		3.00	0.70	3.50

What is the PBV ratio for Firm A?

A) 1.25.



B) 0.71.



C) 2.14.



Explanation

The growth rate in dividends (g) = $ROE(1 - \text{payout ratio}) = 0.20 \times (1 - 0.75) = 0.05$ or 5%.

The PBV ratio = $(ROE - g) / (r - g) = (0.20 - 0.05) / (0.12 - 0.05) = 2.14$.

(Module 22.4, LOS 22.i)

Carol Jenkins, CFA, works as a stock analyst for Cape Cod Partners, a money-management firm that handles private accounts for high net worth clients. Jenkins' assignment is to find attractively valued stocks for client portfolios.

Jenkins believes that recent weakness in the technology sector presents an attractive opportunity. She is looking at Massive Tech, the market leader in chipsets for laptop computers, and Mouse & Associates, a tiny software developer specializing in data-storage programs. Jenkins is considering the companies' relative values in a number of ways. Statistics for Massive and Mouse are provided below:

	Massive Tech	Mouse & Associates
Stock price	\$65	\$12
Trailing earnings	\$4,300	\$3.15
Market capitalization	\$130,000	\$84

Assets	\$16,250	\$7.0
Equity	\$12,000	\$5.5
Operating margin	49%	54%
Net margin	12%	22%
Depreciation	\$3,500	\$6
Amortization	\$5,675	\$1.5
Fixed investment plus borrowing	\$4,200	\$0.3
Dividends	\$3	\$0.02
Shares outstanding	2,000	7

* All figures except stock price, dividends, and percentages are in millions.

In most cases, Jenkins values her stocks relative to an equally-weighted basket of stocks in the same industry in order to avoid significant fundamental differences between companies of different types. However, her picks made based on price/earnings ratios are not doing well against the market. She fears the stocks she selects are not as cheap as she originally thought, relative to her benchmark.

Jenkins also wants to improve Cape Cod's selection of software stocks. To widen the field beyond the companies she currently follows, Jenkins wants to include Canadian software stocks in Cape Cod's research universe. Differences in accounting methodologies are not a concern, but Jenkins is still concerned about the difficulty of valuing the different stocks.

Jenkins has assembled the following data about Canadian software companies:

- Most are very small.
- Most carry little debt, but about 20% are heavily leveraged.
- These companies are more likely to be unprofitable compared to U.S. companies.
- Few pay dividends, as is the case in the U.S.
- Many of the companies are government-subsidized, which leads to drastic differences in the level of operating expenses.

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Question ID: 1473198

Which of the following explanations is *least likely* to explain why Jenkins' stock picks underperform?

- A) Large stocks have an outsized effect on the benchmark data. ✔
- B) She is using the mean rather than the median valuation as a benchmark. ✘
- C) Many stocks in the benchmark group are mispriced. ✘

Explanation

Capitalization weights are not an issue unless the benchmark is a cap-weighted index. Jenkins is using an equally-weighted basket of stocks in the same industry (or simple average). Average valuations reflect outliers; medians do not. P/Es can get very high, but can never fall below zero. As such, the outliers are going to trend high, and the median is likely to be considerably lower than the mean. A stock that looks cheap relative to the mean may look expensive relative to the median. Stocks of different sizes often have different average or median valuations. Mispricing of stocks in the benchmark is always a risk.

(Module 22.4, LOS 22.m)

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Question ID: 1473199

Which valuation ratio is *least* appropriate for comparing Massive and Mouse?

- A) Enterprise value/EBITDA because Massive and Mouse have very different debt levels. 
- B) Price/book because Massive is larger than Mouse. 
- C) Price/cash flow because cash flows for small companies can be extremely volatile. 

Explanation

The P/B ratios can be misleading when used to compare companies with vastly different asset bases. A large semiconductor company is likely to have lots of fixed assets, while a tiny software company may have very few assets. The P/CF ratio tends to be more stable than the P/E ratio. The P/E ratio is useless for considering companies that lose money, but that does not mean the measure has no value when earnings are positive. The EV/EBITDA ratio is effective at comparing stocks with different degrees of financial leverage.

(Module 22.4, LOS 22.m)

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Question ID: 1473200

Mouse & Associates is cheaper than Massive Tech as measured by:

- A) the price/sales ratio and the dividend yield. 
- B) the earnings yield but not the price/book. 
- C) the price/sales ratio and the price/earnings ratio. 

Explanation

To calculate the P/E, divide the market capitalization by the earnings. Lower is cheaper.

To calculate the P/B, divide the market capitalization by the equity. Lower is cheaper.

To calculate the P/S, determine sales by dividing the earnings by the net margin. Then divide the market capitalization by the sales. Lower is cheaper.

To calculate the earnings yield, divide the earnings by the market capitalization. Higher is cheaper.

To calculate the dividend yield, divide the dividends by the price. Higher is cheaper.

	Massive Tech	Mouse & Associates
P/E	30.23	26.67
P/B	10.83	15.27
P/S	3.63	5.87
Earnings yield	3.31%	3.75%
Dividend yield	4.62%	0.17%

(Module 22.4, LOS 22.m)

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Question ID: 1473201

The price/cash flow ratio of Massive Tech, where cash flow is defined as earnings plus noncash charges, is *closest* to:

- A) 16.67.
- B) 9.65.
- C) 7.89.



Explanation

Cash flow = net income plus depreciation plus amortization = (\$4,300 + 3,500 + 5,675) = \$13,475 million.

P/CF = market capitalization/cash flow = (\$130,000/13,475) = 9.65.

(Module 22.4, LOS 22.m)

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Question ID: 1473193

Proprietary Technologies, Inc., (PTI) has a leading price-to-earnings (P/E) ratio of 28 while the median leading P/E of a peer group of companies within the industry is 38. Based on the method of comparables, an analyst would *most likely* conclude that PTI should be:

- A) bought as an undervalued stock.
- B) sold as an overvalued stock.



C) sold short as an overvalued stock.



Explanation

The price per dollar of earnings is considerably lower than that for the median of the peer group, which implies that it may well be undervalued.

(Module 22.4, LOS 22.m)

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Question ID: 1473205

Which of the following is a disadvantage to using EV/EBITDA?

- A) EBITDA is useful for valuing capital-intensive businesses with high levels of depreciation and amortization.
- B) EBITDA is usually positive even when EPS is not.
- C) Since FCFF captures the amount of capital expenditures, it is more strongly linked with valuation theory than EBITDA.



Explanation

Since FCFF captures the amount of capital expenditures, it is more strongly linked with valuation theory than EBITDA. The other statements are advantages.

(Module 22.4, LOS 22.o)

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Question ID: 1473178

At a regional security analysts conference, Sandeep Singh made the following comment: "A PEG ratio is a very useful valuation metric because it generates meaningful results for all equities, regardless of the rate of dividend growth." Is Singh correct?

- A) No, because the PEG ratio generates highly questionable results for low-growth companies.
- B) Yes, because the expected dividend growth rate is cancelled out in the computation of the PEG ratio.
- C) Yes, because the computation of the PEG ratio does include the rate of expected dividend growth.



Explanation

The PEG ratio measures the tradeoff between P/E and expected earnings growth (g). The formula for the PEG ratio is: $PEG = (P/E) / g$. PEG ratios generate questionable results for low-growth companies. Also, the PEG ratio is undefined for companies with zero expected growth (division by zero) or meaningless for companies with negative expected earnings growth.

(Module 22.4, LOS 22.j)

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Question ID: 1473117

One disadvantage of using the price/sales (P/S) multiple for stock valuation is that:

- A) sales are relatively stable and might not change even though earnings and value might change significantly. 
- B) profit margins are not consistent across firms within an industry. 
- C) P/S multiple does not provide a framework to evaluate the effects of corporate policy decisions and price changes. 

Explanation

The stability of sales (relative to earnings and book value) can be a disadvantage. For example, revenues may remain stable but earnings and book values can drop significantly due to a sharp increase in expenses.

(Module 22.3, LOS 22.c)

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Question ID: 1473139

The observation that negative price to earnings (P/E) ratios are meaningless and prices are never negative is used to justify which valuation approach?

- A) Earnings yield. 
- B) Dividend discount model. 
- C) Dividend yield. 

Explanation

The observation is used to justify the earnings yield approach. Negative P/E ratios are meaningless. In such cases, it is common to use normalized earnings per share (EPS) and/or restate the ratio as the earnings yield or E/P because price is never negative. Price to earnings (P/E) ranking can then proceed as usual.

(Module 22.4, LOS 22.f)

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Question ID: 1473218

An analyst is preparing a presentation on "Interpreting PE ratios" and has the following data:

	<i>Portfolio %</i>	<i>Stock PE</i>
Stock AAA	60%	10
Stock BBB	40%	15

Which of the following is the *most appropriate* measure for calculating the portfolio P/E?

- A) Weighted harmonic mean of the P/E's. 
- B) Geometric mean of the P/E's. 

C) Arithmetic average of the P/E's.



Explanation

The weighted harmonic mean of the 10 and 15 will give the result closest to the portfolio earnings divided by the portfolio value.

(Module 22.4, LOS 22.r)

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Question ID: 1473099

The multiple indicated by applying the discounted cash flow (DCF) model to a firm's fundamentals is necessarily the:

A) justified price multiple.



B) result of calculating retention/(required rate of return - growth) for the overall market.



C) same as the average industry multiple.



Explanation

A justified price multiple is the warranted or intrinsic price multiple. It is the estimated fair value of that multiple. The question is limited to an individual firm and does not necessarily apply to the market or an industry.

(Module 22.1, LOS 22.b)

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Question ID: 1473114

The Farmer Co. has a payout ratio of 70% and a return on equity (ROE) of 14%. What will be the appropriate price-to-book value (PBV) based on fundamentals if the expected growth rate in dividends is 4.2% and the required rate of return is 11%?

A) 0.64.



B) 1.44.



C) 1.50.



Explanation

Based on fundamentals:

$$P/BV = (0.14 - 0.042) / (0.11 - 0.042) = 1.44.$$

(Module 22.2, LOS 22.i)

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Question ID: 1473131

At a CFA society function, Robert Chan comments to Li Chiao that the expected dividend growth rate for Xanadu Industries has decreased 0.5% from 6.0% to 5.5%. Chan claims that since Xanadu will maintain their historic dividend payout ratio of 40% and required return on equity (r) of 12%, Xanadu's justified leading P/E ratio based on forecasted fundamentals will also decrease by 0.5%. Is Chan correct?

- A) No, Xanadu's justified leading P/E ratio will decrease by approximately 7.8%. 
- B) No, Xanadu's justified leading P/E ratio will increase by approximately 7.8%. 
- C) Yes, Xanadu's justified leading P/E ratio will increase by approximately 0.5%. 

Explanation

Chan is not correct. $P/E_{\text{Xanadu}} = \text{payout ratio} / (r - g)$

When the expected dividend growth is 6%, $P/E = 0.40 / (0.12 - 0.06) = 6.67$

When the expected dividend growth is 5.5%, $P/E = 0.40 / (0.12 - 0.055) = 6.15$

The percentage change is $(6.15 / 6.67) - 1 = -7.80\%$, representing a 7.80% decrease.

(Module 22.4, LOS 22.d)

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Question ID: 1473125

The following data was available for Morris, Inc., for the year ending December 31, 2001:

- Sales per share = \$150.
- Earnings per share = \$1.75.
- Return on Equity (ROE) = 16%.
- Required rate of return = 12%.

If the expected growth rate in dividends and earning is 4%, what will the appropriate price-to-sales (P/S) multiple be for Morris?

- A) 0.114. 
- B) 0.109. 
- C) 0.037. 

Explanation

Profit Margin = $EPS / \text{Sales per share} = 1.75 / 150 = 0.01167$ or 1.167%.

Payout ratio = $1 - (g / ROE) = 1 - (0.04 / 0.16) = 0.75$ or 75%.

$P_0 / S_0 = [\text{profit margin} \times \text{payout ratio} \times (1 + g)] / (r - g) = [0.01167 \times 0.75 \times 1.04] / (0.12 - 0.04) = 0.11375$.

(Module 22.3, LOS 22.i)

Question #19 of 140

Question ID: 1473204

Which of the following measures of cash flow is *most closely* linked with valuation theory?

- A)** Earnings before interest, taxes, depreciation, and amortization (EBITDA). 
- B)** Free cash flow to equity (FCFE). 
- C)** Cash flow from operations (CFO). 

Explanation

FCFE is most strongly linked to valuation theory. Both remaining proxies are in need of significant adjustment to accurately measure cash flow in valuation.

(Module 22.4, LOS 22.n)

Question #20 of 140

Question ID: 1473153

The Lewis Corp. had revenue per share of \$300 in 2001, earnings per share of \$4.50, and paid out 60% of its earnings as dividends. If the return on equity (ROE) and required rate of return of Lewis are 20% and 13% respectively, what is the appropriate price/sales (P/S) multiple for Lewis?

- A)** 0.12. 
- B)** 0.18. 
- C)** 0.19. 

Explanation

Profit Margin = EPS / Sales per share = 4.50 / 300 = 0.015 or 1.5%.

Expected growth in dividends and earnings = ROE × (1 – payout ratio) = 0.20 × 0.40 = 0.08 or 8%.

$P_0/S_0 = [\text{profit margin} \times \text{payout ratio} \times (1 + g)] / (r - g) = [0.015 \times 0.60 \times (1.08)] / (0.13 - 0.08) = 0.1944$.

(Module 22.4, LOS 22.i)

An analyst has gathered the following fundamental data:

	Firm A	Firm A	Firm B	Firm B
Strategy	High Margin Low Volume	Low Margin High Volume	High Margin Low Volume	Low Margin High Volume
Payout Ratio	40%	40%	40%	40%
Required Rate of Return	11%	11%	11%	11%
Growth Rate in Dividends	9%	5%	5%	7%
Sales/Book Value of Equity	1.5	4.5	1.0	3
Profit Margin	10%	2%	9%	4%
Book Value	\$150	\$150	\$125	\$125

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Question ID: 1586170

What is the price-to-sales (P/S) multiple for Firm A in the high-margin, low-volume strategy?

- A) 2.18. 
- B) 0.13. 
- C) 2.00. 

Explanation

The P/S multiple = $[\text{Profit Margin} \times \text{Payout Ratio} \times (1 + g)] / (r - g) = (0.10 \times 0.4 \times 1.09) / (0.11 - 0.09) = 2.18$.

(Module 22.2, LOS 22.i)

Question #22 - 22 of 140

Question ID: 1586171

What is the P/S multiple for Firm B in the low-margin, high-volume strategy?

- A) 0.43. 
- B) 0.60. 
- C) 2.00. 

Explanation

The P/S multiple = $[\text{Profit Margin} \times \text{Payout Ratio} \times (1 + g)] / (r - g) = (0.04 \times 0.4 \times 1.07) / (0.11 - 0.07) = 0.428$ or 0.43.

(Module 22.2, LOS 22.i)

Question #23 of 140

Question ID: 1473135

Underlying earnings may be defined as earnings:

- A) net of capital expenditures needed to keep the business productive. 
- B) that include non-recurring components. 
- C) that exclude non-recurring components. 

Explanation

Underlying earnings are earnings that exclude non-recurring items. They are also known as persistent, continuing, or core earnings.

(Module 22.4, LOS 22.e)

Question #24 of 140

Question ID: 1473162

A firm's return on equity (ROE) is 15%, its required rate of return is 12%, and its expected growth rate is 7%. What is the firm's justified price to book value (P/B) based on these fundamentals?

A) 1.60.



B) 0.63.



C) 1.71.



Explanation

$$P_0/B_0 = (ROE - g) / (r - g) = (0.15 - 0.07) / (0.12 - 0.07) = 1.60$$

(Module 22.4, LOS 22.i)

Question #25 of 140

Question ID: 1473173

For which of the following firms is the Price/Earnings to Growth (PEG) ratio *most* appropriate for identifying undervalued or overvalued equities?

Firm A: Expected dividend growth = 6%; Cost of equity = 12%; price-to-earnings (P/E) = 12.

Firm B: Expected dividend growth = -6%; Cost of equity = 12%; price-to-earnings (P/E) = 12.

Firm C: Expected dividend growth = 1%; Cost of equity = 12%; price-to-earnings (P/E) = 12.

A) Firm B.



B) Firm A.



C) Firm C.



Explanation

The formula for the PEG ratio is: $PEG = (P/E) / g$. It measures the tradeoff between P/E and expected dividend growth (g). For traditional growth firms, PEG ratios fall between 1 and 2. The general rule is that PEG ratios above 2 are indicative of overvalued firms (expensive), and PEG ratios below 1 are indicative of firms that are undervalued (cheap).

Firm A:	PEG = 2, indicating a stock that is appropriately priced.
Firm B:	The PEG ratio of firms with negative expected dividend growth is negative, which is meaningless. For Firm B, PEG = -2.
Firm C:	Firms with very low expected dividend growth are likely to have PEG ratios that unrealistically indicate overvalued stocks. For Firm C, PEG = 12.

(Module 22.4, LOS 22.j)

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Question ID: 1473089

Shares of TKR Construction (TKR) are selling for \$50. Earnings for the last 12 months were \$4.00 per share. The average trailing P/E ratio for firms in TKR's industry is 15. The appropriate WACC is 12%, and the risk-free rate is 8%. Assume a growth rate of 0%. Using the method of comparables, what price is indicated for TKR?

A) \$50.00.



B) \$60.00.



C) \$33.33.



Explanation

Using the method of comparables, TKR should be priced as $(15 \times 4) = \$60.00$.

(Module 22.1, LOS 22.a)

Question #27 of 140

Question ID: 1473206

An analyst gathers the following information for ABC Industries:

Market Value of Debt	\$110 million
Market Value of Equity	\$90 million
Book Value of Debt	\$100 million
Book Value of Equity	\$50 million
EBITDA	\$75 million

The EV/EBITDA is *closest* to:

A) 2.00.



B) 2.67.



C) 2.13.



Explanation

EV uses market values for debt and equity. $(110 + 90) / 75 = 2.67$.

(Module 22.4, LOS 22.o)

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Question ID: 1473110

The price-to-book value (PBV) ratio for a high-growth firm will:

A) increase as the growth rate in either the high-growth or stable-growth period increases.



B) increase as the growth rate in the high-growth period increases and decrease as the growth rate in the stable-growth period increases.



C) increase as the growth rate in either the high-growth or stable-growth period decreases.



Explanation

The PBV ratio for a high-growth firm will be determined by growth rates in earnings in both the high-growth and stable-growth periods. The PBV ratio increases as the growth rate increases in either period.

(Module 22.2, LOS 22.g)

Question #29 of 140

Question ID: 1473159

A firm has a payout ratio of 40%, a profit margin of 7%, an estimated growth rate of 10%, and its shareholders require a return of 14% on their investment. Based on these fundamentals, a reasonable estimate of the appropriate price-to-sales ratio for the firm (based on trailing sales) is:

- A) 0.56. ✗
- B) 0.77. ✓
- C) 0.70. ✗

Explanation

$$\frac{P}{S} = \frac{\text{payout} \times \text{profitmargin} \times (1 + g)}{r - g} = \frac{0.40 \times 0.07 \times 1.10}{0.14 - 0.10} = 0.77$$

(Module 22.4, LOS 22.i)

Question #30 of 140

Question ID: 1473154

What is the appropriate justified trailing price-to-earnings (P/E) multiple of a stock that has a payout ratio of 40% if shareholders require a return of 15% on their investment and the expected growth rate in dividends is 5%?

- A) 6.30. ✗
- B) 4.20. ✓
- C) 3.80. ✗

Explanation

$$P_0/E_0 = (0.40 \times 1.05) / (0.15 - 0.05) = 4.20$$

(Module 22.4, LOS 22.i)

Question #31 of 140

Question ID: 1508668

An *increase* in which of the following variables will *least likely* result in a corresponding *increase* in the price-to-book value (PBV) ratio for a high-growth firm?

- A) Payout ratios. ✗

B) Required rate of return.



C) Growth rates in earnings.



Explanation

The PBV ratio decreases as the required rate of return increases.

(Module 22.4, LOS 22.g)

Question #32 of 140

Question ID: 1473194

Enhanced Systems, Inc., has a price to book value (P/B) of five while the median P/B of a peer group of companies within the industry is five. Based on the method of comparables, an analyst would *most likely* conclude that ESI should be:

A) sold or sold short as an overvalued stock.



B) bought as an undervalued stock.



C) viewed as a properly valued stock.



Explanation

The price per dollar of book value is the same as that for the median of the peer group, which implies that it is likely properly valued.

(Module 22.4, LOS 22.m)

Question #33 of 140

Question ID: 1473217

An analyst is preparing a presentation on "Interpreting PE ratios" and has the following data:

	<i>Portfolio %</i>	<i>Stock PE</i>
Stock AAA	60%	10
Stock BBB	40%	15

Which of the following is *closest* to the weighted harmonic mean of these two PE ratios?

A) 11.98.



B) 12.49.



C) 11.54.



Explanation

The weighted harmonic mean of the two PE ratios is a harmonic mean which is weighted by the portfolio weights.

$$1/[(0.60 \times 1/10) + (0.40 \times 1/15)] = 11.54$$

(Module 22.4, LOS 22.r)

Beyan Bautista, CFA, is a sell-side research analyst for a boutique UK investment house. One of the companies she is currently covering is Yantra Plc, a manufacturer of mid-range motorboats, primarily for river use. The company's shares closed at £44.56 on 15 January 20x4. Beyan uses the Treasury bond yield of 2.5% and an assumed market risk premium of 5% when calculating justified ratios based on forecasted fundamentals.

She would first like to calculate Yantra's normalized price-to-earnings ratio over the period 20x0 to 20x3 via the following two methods:

- Method 1 – historical average EPS
- Method 2 – average return on equity

The following table summarizes selected historical data for Yantra Plc. All GBP figures are quoted on a per-share basis:

	20x0	20x1	20x2	20x3
Sales				£32.44
Earnings	£1.57	£2.16	£3.24	£1.89
Dividends				£0.76
FCFE				£1.05
Book value	£10.56	£11.88	£14.21	£15.69
ROE	14.9%	18.2%	22.8%	12.0%
Beta				1.40

Beyan would like to value Yantra using justified trailing price-to-sales and price-to-book ratios based on forecasted fundamentals. She bases the inputs on the most recent data (i.e., 20x3).

Finally, she would like to conclude her valuation by analyzing the justified trailing price- to-cash flow and justified trailing dividend yield metrics. As above, where needed, 20x3 data is used to develop the inputs. Beyan assumes cash flows are growing at a constant rate.

Given the recent lukewarm reception of a new product line at Yantra geared at ocean use, Beyan is considering increasing the cost of equity and reducing the growth rate in her models. She is considering what impact this might have on justified ratios based on forecasted fundamentals such as P/E, P/B, P/S, P/CF, D/P.

Beyan would also like to apply multiples analysis to three of Yantra's closest competitors—Arda, Struma, and Tundzha. Having read the MD&A of the annual reports of each company, she has highlighted the following points:

Arda: "is in financial hardship due to a recent downturn in its sector"

Struma: "operates in extremely cyclical industry"

Tundzha: "has very different cost structure to the rest of its peers"

Based on the highlighted points, she is deciding on the most appropriate candidate ratio for each of the above companies:

Arda: price-to-earnings or price-to-book

Struma: normalized price-to-earnings or trailing price-to-sales

Tundzha: price-to-earnings or price-to-sales

Beyan has also been recently tasked with covering Nanuk Plc and Nunca Plc, two close competitors developing innovative solutions for marine navigation. She has collected the following information on the two companies (the trailing cash flow per share is calculated as net income per share plus non-cash charges per share):

	Nanuk	Nunca
Share price	38.00	64.00
Trailing CF per share	4.52	9.11
P/CF	8.41	7.03
Trailing FCFE per share	3.11	3.85
P/FCFE	12.22	16.62
5-year growth rate	14%	19%
Beta	1.4	1.4

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Question ID: 1473166

Using the P/E ratio with normalized earnings, Yantra appears to be more attractively valued under:

- A) Method 1.
- B) Method 2.
- C) Neither method as they result in the same conclusion.



Explanation

Method 1:

Calculate average EPS over the stated period. $(1.57 + 2.16 + 3.24 + 1.89) / 4 = £2.22$.

Based on a price of £44.56, the P/E multiple is $44.56 / 2.22 = 20.12$

Method 2:

First, calculate average ROE over the period

$(14.9\% + 18.2\% + 22.8\% + 12\%) / 4 = 17\%$.

To get the normalized earnings multiply the average ROE by the most recent book value of £15.69 – $17\% \times 15.69 = £2.66$. Thus, the resulting P/E multiple for Method 2 is $44.56 / 2.66 = 16.73$.

Comparing the two multiples, Method 2 results in higher normalized earnings and, therefore, lower P/E ratio and hence a lower (more attractive) valuation.

(Module 22.4, LOS 22.e)

Question #35 - 39 of 140

Question ID: 1586185

Using justified trailing price-to-sales and price-to-book ratios based on forecasted fundamentals, Yantra appears to be:

A) undervalued.



B) overvalued.



C) the results are mixed.



Explanation

The price-to-sales multiple based on market data is $44.56 / 32.44 = 1.37$

The price-to-book multiple based on market data is $44.56 / 15.69 = 2.84$.

The formula for justified price-to-sales ratio based on forecasted fundamentals is:

$$\frac{P_0}{S_0} = \frac{PM \times (1-b) \times (1+g)}{(r-g)}$$

The profit margin (PM) is $20X3 \text{ EPS} / \text{sales per share} = 1.89 / 32.44 = 5.8\%$

The dividend-payout-ratio $(1 - b)$ is $0.76 / 1.89 = 40\%$.

The retention rate (b) is 60% .

The sustainable growth rate is $ROE \times b = 12\% \times 60\% = 7.2\%$

The cost of equity from CAPM is $2.5\% + 1.4 \times 5\% = 9.5\%$

Thus, the justified price-to-sales ratio is:

$$\frac{P_0}{S_0} = \frac{0.058 \times 0.40 \times 1.072}{(0.095 - 0.072)} = 1.08$$

Comparing the justified multiple of 1.09 to the market based multiple of 1.37 the company appears **overvalued**.

The formula for justified price-to-book ratio based on forecasted fundamentals is:

$$\frac{P_0}{B_0} = \frac{(ROE-g)}{(r-g)} = \frac{(0.12-0.072)}{(0.095-0.072)} = 2.11$$

Comparing the justified multiple of 2.11 to the market based multiple of 2.84 the company, again, appears **overvalued**.

Please, note that the value of each ratio is not required in the solution. Therefore, realizing that justified ratios are derivable from the Gordon Growth Model (GGM), one could simply proceed and value the company as follow:

$$P_0 = \frac{D_0 \times (1+g)}{(r-g)} = \frac{0.76 \times (1+0.072)}{(0.095-0.072)} = 35.42$$

The company is overvalued as its market price of £44.56 exceeds its intrinsic value of £35.42. As a result, any ratio that divides this intrinsic value by a value driver (e.g., P/B, P/S, etc.) will also be higher than the market multiple resulting in the same conclusion.

(Module 22.2, LOS 22.i)

Question #36 - 39 of 140

Question ID: 1586186

Using justified trailing price-to-cash flow ratio and dividend yield based on forecasted fundamentals, Yantra appears to be:

A) undervalued.



B) overvalued.



C) the results are mixed.



Explanation

The price-to-cash flow multiple based on market data is $44.56 / 1.05 = 42.44$.

The dividend yield flow multiple based on market data is $0.76 / 44.56 = 1.7\%$.

The intrinsic value of a company based on FCFE and the Gordon Growth Model is:

$$V_0 = \frac{FCFE_0 \times (1+g)}{(r-g)}$$

To obtain the trailing justified price-to-cash flow; divide both sides of the equation by FCFE₀. We already have g and r inputs from prior computations:

$$\frac{P}{CF} = \frac{(1+g)}{(r-g)} = \frac{1.072}{(0.095-0.072)} = 46.65$$

Thus, the company appears **undervalued** based on this criterion as its market multiple of 42.44 is below the justified multiple of 46.65.

Reciprocating the Gordon Growth Model, we obtain the formula for justified trailing dividend yield:

$$\frac{D_0}{P_0} = \frac{(r-g)}{(1+g)}$$

This is the reciprocal of the above P/CF ratio, so a shortcut computation would be:

$$\text{justified } D/P = 1 / 46.65 = 2.1\%$$

The current dividend yield of 1.7% is below the justified dividend yield of 2.1% so, based on this criterion, the company appears **overvalued** (note that when price is on the numerator high multiple = undervalued).

(Module 22.2, LOS 22.i)

Question #37 - 39 of 140

Question ID: 1586187

If the appropriate adjustments to the five justified ratios are implemented following the launch of the new product line at Yantra, then:

- A) all five ratios will decline.
- B) four ratios will decline.
- C) three ratios will decline.



Explanation

Generally, most justified ratios suffer when the discount rate is increased and/or assumed growth rate decreased. However, the notable exception to this rule is justified dividend yield. Lower growth implies more earnings available for dividend payments. Higher cost of equity reduces share price, thus (maintaining a constant dollar dividend) the dividend yield increases.

(Module 22.2, LOS 22.g)

In relation to Arda, Struma, and Tundzha, Beyan should opt for:

	<u>Arda</u>	<u>Struma</u>	<u>Tundzha</u>	
A)	P/B	norm P/E	P/E	✓
B)	P/B	norm P/E	P/S	✗
C)	P/E	P/S	P/E	✗

Explanation

Arda is in financial hardship, which probably means the company has very low or even negative earnings rendering the P/E ratio meaningless.

Struma operates in a very cyclical industry so earnings normalization is necessary to take into account the full impact of the business cycle. Taking the trailing price-to-sales ratio, (i.e., most recent twelve-month sales) would either inflate or deflate the ratio for a cyclical company depending on the stage of the cycle.

Tundzha has a very different cost structure relative to its peer group. This indicates that the use of the price-to-sales ratio is not a good idea as that ratio completely ignores items below the sales line (i.e., ignores cross-sectional differences in profitability).

(Module 22.1, LOS 22.c)

Question #39 - 39 of 140

Question ID: 1473171

In relation to the companies in the marine navigation sector, Nunca is:

- A) undervalued relative to Nanuk. ✗
- B) overvalued relative to Nanuk. ✗
- C) trading at premium due to its superior fundamentals. ✓

Explanation

First, note that computing "cash flow" as net income plus non-cash charges is suboptimal and should not be trusted; especially if a superior metric such as price-to-FCFE is present.

Therefore, despite the fact, Nunca has a lower price-to-cash flow ratio; this is unlikely to be a reason to invest, especially as it is in contradiction to the superior price-to-FCFE ratio.

Controlling for risk (i.e., companies have the same beta), we note that Nunca has a higher five-year estimated growth rate but also higher P/FCFE multiple. The company is therefore not necessarily either under or overvalued relative to its competitor (based on the limited information presented in the table), but it is certainly trading at a premium (i.e., trading at a higher multiple) due to its higher growth forecast.

(Module 22.1, LOS 22.c)

Question #40 of 140

Question ID: 1473138

The average return on equity (ROE) earnings normalization method relies on:

- A) average ROE over the most recent cycle. 
- B) the earnings yield. 
- C) average earnings per share (EPS) over the most recent cycle. 

Explanation

The average return on equity normalization method normalizes EPS as the average ROE over the most recent full cycle multiplied by book value per share.

(Module 22.4, LOS 22.e)

Question #41 of 140

Question ID: 1473146

An analyst is valuing a company with a dividend payout ratio of 0.35, a beta of 1.45, and an expected earnings growth rate of 0.08. A regression on comparable companies produces the following equation:

Predicted price to earnings (P/E) = $7.65 + (3.75 \times \text{dividend payout}) + (15.35 \times \text{growth}) - (0.70 \times \text{beta})$

What is the predicted P/E using the above regression?

- A) 11.21. 
- B) 9.18. 
- C) 7.65. 

Explanation

Predicted P/E = $7.65 + (3.75 \times 0.35) + (15.35 \times 0.08) - (0.70 \times 1.45) = 9.1755$

(Module 22.4, LOS 22.h)

Question #42 of 140

Question ID: 1473145

An analyst is valuing a company with a dividend payout ratio of 0.55, a beta of 0.92, and an expected earnings growth rate of 0.07. A regression on comparable companies produces the following equation:

Predicted price to earnings (P/E) = $7.65 + (3.75 \times \text{dividend payout}) + (15.35 \times \text{growth}) - (0.70 \times \text{beta})$

What is the predicted P/E using the above regression?

- A) 11.43. 
- B) 7.65. 
- C) 10.14. 

Explanation

Predicted P/E = $7.65 + (3.75 \times 0.55) + (15.35 \times 0.07) - (0.70 \times 0.92) = 10.14$

(Module 22.4, LOS 22.h)

Question #43 of 140

Question ID: 1586175

Leslie Singer comments to Robert Chan that Dreamtime Industries' expected dividend growth rate is 5.0%, ROE is 14%, and required return on equity (r) is 10%. Based on a justified P/B ratio compared to a P/B ratio (based on market price per share) of 1.60, Dreamtime Industries is *most likely*:

A) undervalued.



B) correctly valued.



C) overvalued.



Explanation

Justified P/B = $(\text{ROE} - g) / (r - g)$. When the expected dividend growth is 5.0%, the justified P/B = $(0.14 - 0.05) / (0.10 - 0.05) = 1.80$. This is greater than the market P/B of 1.60.

(Module 22.2, LOS 22.I)

Question #44 of 140

Question ID: 1586172

What is the justified leading price-to-earnings (P/E) multiple of a stock that has a retention ratio of 60% if the shareholders require a return of 16% on their investment and the expected growth rate in dividends is 6%?

A) 4.24.



B) 6.36.



C) 4.00.



Explanation

Justified Leading P/E = $P_0/E_1 = 1 - b / r - g = \text{Payout ratio} / r - g = 0.40 / (0.16 - 0.06) = 4.00$

(Module 22.2, LOS 22.i)

Question #45 of 140

Question ID: 1473116

An argument against using the price-to-sales (P/S) valuation approach is that:

A) P/S ratios are not as volatile as price-to-earnings (P/E) multiples.



B) P/S ratios do not express differences in cost structures across companies.



- C) sales figures are not as easy to manipulate or distort as earnings per share (EPS) and book value. 

Explanation

P/S ratios do not express differences in cost structures across companies. Both remaining responses are advantages of the P/S ratios, not disadvantages.

(Module 22.3, LOS 22.c)

Question #46 of 140

Question ID: 1473133

The goal of normalizing earnings is to adjust for:

- A) cyclical elements. 
- B) seasonal elements. 
- C) non-cash charges. 

Explanation

The goal of normalizing earnings is to adjust for cyclical elements.

(Module 22.4, LOS 22.e)

Question #47 of 140

Question ID: 1473132

Which of the following statements about cyclical firms is *least* accurate?

- The problems encountered when using the price-to-earnings (P/E) multiples of
- A) cyclical firms can be completely eliminated by using average or normalized earnings. 
- B) Cyclical firms have volatile earnings, and their price-to-earnings (P/E) multiple is not very useful for valuation. 
- C) The price-to-earnings (P/E) multiple of a cyclical firm normally peaks at the depths of recession and bottoms out at the peak of economic boom. 

Explanation

The P/E multiples for cyclical firms are not very useful for valuation. Earnings will follow the economy, and prices will reflect expectations about the future. Thus, most of the time, the P/E multiple of a cyclical firm will peak at the depths of recession and bottom out at the peak of an economic boom. This problem can be minimized to some extent by using average or normalized earnings but will not be eliminated completely.

(Module 22.4, LOS 22.e)

Question #48 of 140

Question ID: 1473202

Earnings before interest, taxes, depreciation, and amortization (EBITDA) is *best* suited as a measure of:

- A) debt capacity. 
- B) equity value. 
- C) total company value. 

Explanation

EBITDA is a pre-tax, pre-interest measure, which represents a flow to both equity and debt. Thus, it is better suited as an indicator of total company value than just equity value.

(Module 22.4, LOS 22.n)

Question #49 of 140

Question ID: 1473097

A justified price multiple is the:

- A) multiple implied by the market price. 
- B) multiple implied by historical growth. 
- C) warranted or intrinsic price multiple. 

Explanation

A justified price multiple is the warranted or intrinsic price multiple. It is the estimated fair value of that multiple.

(Module 22.1, LOS 22.b)

Question #50 of 140

Question ID: 1473208

Which of the following are advantages of using EV/EBITDA?

- A) EV/EBITDA ignores how different revenue recognition policies affect CFO. 
- B) EBITDA is useful for valuing capital-intensive businesses with high levels of depreciation and amortization. 
- C) If working capital is growing, EBITDA will be larger than CFO. 

Explanation

EBITDA is useful for valuing capital-intensive businesses with high levels of depreciation and amortization. The other statements are disadvantages to using EV/EBITDA.

(Module 22.4, LOS 22.o)

Question #51 of 140

Question ID: 1473120

An increase in growth will cause a price to cash flow multiple to:

A) there is insufficient information to tell.



B) decrease.



C) increase.



Explanation

An increase in growth increases the price to cash flow ratio (CF), as indicated by the following expression:

$$P_0 / CF_0 = (1 + g) / (r - g)$$

(Module 22.3, LOS 22.g)

Question #52 of 140

Question ID: 1473111

The net impact of an *increase* in payout ratio on price-to-book value (PBV) ratio cannot be determined because it might also:

A) decrease required rate of return.



B) decrease expected growth.



C) decrease the market value of the firm.



Explanation

If payout increases, the growth of the firm may slow down, because internally generated funds are not being invested in new, profitable projects. Hence, the net impact on the PBV ratio from change in payout ratio cannot be determined.

(Module 22.2, LOS 22.g)

Question #53 of 140

Question ID: 1473157

Margin and Sales Trade-off for CVR, Inc. and Home, Inc., for Next Year				
Firm	Strategy	Retention Rate	Profit Margin	Sales/Book Value of Equity
CVR, Inc.	High Margin / Low Volume	20%	8%	1.25
CVR, Inc.	Low Margin / High Volume	20%	2%	4.00
Home, Inc.	High Margin / Low Volume	40%	9%	2.00
Home, Inc.	Low Margin / High Volume	40%	1%	20.0

(Note: CVR, Inc., has a book value of equity of \$80 and a required rate of return of 10%. Home, Inc., has a book value of equity of \$100 and a required rate of return of 11%.)

If CVR, Inc., has a required return for shareholders of 10%, what is its appropriate leading price-to-sales (P/S) multiple if the firm undertakes the high margin/low volume strategy?

- A) 1.46. 
- B) 0.80. 
- C) 0.20. 

Explanation

$$g = \text{Retention Rate} \times \text{Profit Margin} \times \text{Sales/book value of equity} = 0.20 \times 0.08 \times 1.25 = 0.02.$$

If profit margin is based on the expected earnings next period,

$$\text{Leading P/S} = (\text{profit margin} \times \text{payout ratio}) / (r - g) = (0.08 \times 0.80) / (0.10 - 0.02) = 0.80.$$

(Module 22.4, LOS 22.i)

Question #54 of 140

Question ID: 1473106

Which of the following is NOT an advantage of using price-to-book value (PBV) multiples in stock valuation?

- A) Book value is often positive, even when earnings are negative. 
- B) PBV ratios can be compared across similar firms if accounting standards are consistent. 
- C) Book values are very meaningful for firms in service industries. 

Explanation

Book values are NOT very meaningful for firms in service industries.

(Module 22.2, LOS 22.c)

Question #55 of 140

Question ID: 1473160

Margin and Sales Trade-off for CVR, Inc. and Home, Inc., for Next Year				
Firm	Strategy	Retention Rate	Profit Margin	Sales/Book Value (SBV) of Equity
CVR, Inc.	High Margin / Low Volume	20%	8%	1.25
CVR, Inc.	Low Margin / High Volume	20%	2%	4.00
Home, Inc.	High Margin / Low Volume	40%	9%	2.00
Home, Inc.	Low Margin / High Volume	40%	1%	20.0

Note: CVR, Inc., has a book value of equity of \$80 and a required rate of return of 10%.

Home, Inc., has a book value of equity of \$100 and a required rate of return of 11%.

If Home, Inc., has a required return for shareholders of 11%, what is its appropriate leading price-to-sales (P_0 / S_1) multiple if the firm undertakes the low margin/high volume strategy?

- A) 0.80. ✗
- B) 0.20. ✓
- C) 1.00. ✗

Explanation

$$g = \text{Retention Rate} \times \text{Profit Margin} \times \text{SBV of equity} = 0.40 \times 0.01 \times 20.0 = 0.08.$$

If profit margin is based on the expected earnings next period,

$$P/S = (\text{profit margin} \times \text{payout ratio}) / (r - g) = (0.01 \times 0.60) / (0.11 - 0.08) = 0.20.$$

(Module 22.4, LOS 22.i)

Question #56 of 140

Question ID: 1586177

At a CFA society function, Robert Chan comments to Li Chiao that Xanadu Industries' expected dividend growth rate is 5.5%, dividend payout ratio (g) is 40%, and required return on equity (r) is 12%. Based on a justified leading P/E ratio compared to an actual P/E ratio of 8.0, Xanadu Industries is *most likely*:

- A) correctly valued. ✗
- B) overvalued. ✓
- C) undervalued. ✗

Explanation

Justified Leading P/E = payout ratio / ($r - g$). When the expected dividend growth is 5.5%, the justified leading P/E = $0.40 / (0.12 - 0.055) = 6.15$. This is less than the actual (based on current market price) P/E of 8.0.

(Module 22.2, LOS 22.I)

Question #57 of 140

Question ID: 1473163

Industrial Light had earnings per share (EPS) of \$5.00 past year, a dividend per share of \$2.50, a cost of equity of 12%, and a long-term expected growth rate of 5%. What is the trailing price-to-earnings (P/E) ratio?

A) 3.75.



B) 7.50.



C) 7.14.



Explanation

$$PE = \frac{(1-b) \times (1+g)}{r-g}$$

$$1 - b = 1 - (2.50/5.00) = 0.50$$

$$P_5 / E_5 = (0.50 \times 1.05) / (0.12 - 0.05) = 7.50$$

(Module 22.4, LOS 22.i)

Question #58 of 140

Question ID: 1473092

Which of the following valuation approaches is based on the rationale that stock values differ due to differences in the expected values of variables such as sales, earnings, or related growth rates?

A) Method of forecasted fundamentals.



B) Free cash flow to the firm.



C) Method of comparables.



Explanation

The method of forecasted fundamentals is based on the rationale that stock values differ due to differences in the expected values of fundamentals such as sales, earnings, or related growth rates.

(Module 22.1, LOS 22.a)

Beachwood Builders merged with Country Point Homes on December 31, 2003. Both companies were builders of mid-scale and luxury homes in their respective markets. On December 31, 2013, because of tax considerations and the need to segment the businesses between mid-scale and luxury homes, Beachwood

decided to spin-off Country Point, its luxury home subsidiary, to its common shareholders. Beachwood retained Bernheim Securities to value the spin-off of Country Point to its shareholders.

The following information is available to Bernheim's investment bankers:

- Country Point's allocated common equity was \$55.6 million as of December 31, 2013.
- Beachwood paid no dividends and has no preferred shareholders.
- Country Point's free cash flow (FCF) is expected to grow 7% after 2017.
- The current risk-free rate is 6%. The market risk premium is 11%.
- Beachwood Builders had 5 million common shares as of December 31, 2013.
- Country Point's cost of capital is equal to its return on equity at year-end (rounded to the nearest percentage point).
- Country Point did not have any long-term debt allocated from Beachwood.

The following data for Country Point is also available for analysis:

\$ (in millions)	2013	2014(E)	2015(E)	2016(E)	2017(E)
Net Income	10	15	20	25	30
Depreciation	5	6	5	6	5
Change in Capital Expenditures	7	8	9	10	12
Change in Working Capital	0	0	0	0	0

There are three comparable companies in Country Point's peer group: Upscale Homes, Custom Estates and Chateau One.

Company	Forward P/E	Five-Year EPS Growth Forecast	Forward PEG
Upscale Homes	10.0	12.5%	0.80
Custom Estates	15.0	15.0%	1.00
Chateau One	20.0	17.5%	1.14

Question #59 - 62 of 140

Question ID: 1586180

Bernheim's investment bankers have determined the value of Country Point to be \$162.6 million. As part of the spin-off, Beachwood issued to its common shareholders two shares in Country Point for each Beachwood share that its current shareholders held. The appropriate initial offering price per share of the shares that Beachwood's shareholders receive is *closest* to:

- A) \$16.26.
- B) \$32.50.
- C) \$14.45.



Explanation

Since the shareholders receive two shares for every share they currently hold, each Beachwood common shareholder will receive two common shares of Country Point. At December 31, 2013, Beachwood had 5 million shares. Therefore, 10 million common shares were issued for the spin-off. The spin-off was valued at \$162.6 million; dividing by 10 million, we arrive at a spin-off value per share of \$16.26 (= \$162.6 million / 10 million).

(Module 22.4, LOS 22.c)

Question #60 - 62 of 140

Question ID: 1586181

Immediately after the spin-off, Country Point's book value per share is *closest* to:

A) \$16.25.



B) \$5.56.



C) \$11.12.



Explanation

The allocated common equity or book value of Country Point was \$55.6 million at year-end 2013 and 10 million shares were allocated for the spin-off. The book value would be \$55.6 million / 10 million = \$5.56 per share.

(Module 22.4, LOS 22.c)

Question #61 - 62 of 140

Question ID: 1586182

Assume for this question that the initial offering price per share of the Country Point shares is \$16.26. Based on this initial offering price of the spin-off, the estimated price-to-book (P/B) ratio of Country Point is *closest* to:

A) 2.00 times.



B) 1.46 times.



C) 2.92 times.



Explanation

The P/B ratio is determined by taking the spin-off price and dividing it by the book value per share (BVPS). Hence, the ratio is \$16.26 per share spin-off price / \$5.56 BVPS = 2.92 × book.

(Module 22.4, LOS 22.c)

Question #62 - 62 of 140

Question ID: 1586183

Based on Bernheim's careful analysis, firms comparable to Country Point trade at a P/B ratio of 3.5 times. The expected price per share of the spin-off based on this P/B ratio and assuming a liquid and efficient market for Country Point's common shares is *closest* to:

A) \$38.92.



B) \$19.46.



C) \$56.88.



Explanation

Based on the comparable P/B ratio of 3.5 times, we can simply multiply the book value of \$5.56 by 3.5 to arrive at \$19.46.

(Module 22.4, LOS 22.c)

Question #63 of 140

Question ID: 1473118

Which of the following is a disadvantage of using price-to-sales (P/S) multiples in stock valuations?

A) It is difficult to capture the effects of changes in pricing policies using P/S ratios.



B) The use of P/S multiples can miss problems associated with cost control.



C) P/S multiples are more volatile than price-to-earnings (P/E) multiples.



Explanation

Due to the stability of using sales relative to earnings in the P/S multiple, an analyst may miss problems of troubled firms concerning its cost control. P/S multiples are actually less volatile than P/E ratios, which is an advantage in using the P/S multiple. Also, P/S ratios provide a useful framework for evaluating effects of pricing changes on firm value.

(Module 22.3, LOS 22.c)

Question #64 of 140

Question ID: 1473102

An argument against using the price-to-earnings (P/E) valuation approach is that:

A) earnings power is the primary determinant of investment value.



B) research shows that P/E differences are significantly related to long-run average stock returns.



C) earnings can be negative.



Explanation

Negative earnings render the P/E ratio useless. Both remaining factors increase the usefulness of the P/E approach.

(Module 22.1, LOS 22.c)

Question #65 of 140

Question ID: 1473209

An analyst gathered the following data for TRK Construction [all amounts in Swiss francs (Sf)]:

Recent share price	Sf 30.00
Shares outstanding	Sf 40 million
Market value of debt	Sf 120 million
Cash and marketable securities	Sf 75 million
Investments	Sf 200 million
Net income	Sf 160 million
Interest expense	Sf 9 million
Depreciation and amortization	Sf 12 million
Taxes	Sf 48 million

The EV/EBITDA multiple for TRK Construction is *closest* to:

- A) 3.47x.
- B) 5.21x.
- C) 4.56x.



Explanation

EBITDA = (net income + interest + taxes + depreciation / amortization)

EV = (market value of common stock + market value of debt – cash and investments)

EBITDA = 160 + 9 + 12 + 48 = Sf 229 million

EV = (30 × 40) + 120 – 75 – 200 = Sf 1045 million

EV / EBITDA = 4.56

(Module 22.4, LOS 22.o)

Question #66 of 140

Question ID: 1473216

Which of the following is NOT a common momentum valuation indicator?

- A) Dividend yield.
- B) Relative strength.
- C) Earnings surprise.



Explanation

Dividend yield is not generally considered a momentum valuation indicator.

(Module 22.4, LOS 22.q)

Alpha Software (AS) recently reported annual earnings per share (EPS) of \$1.75, which included an extraordinary loss of \$0.19 and an expense of \$0.10 related to acquisition costs during the accounting period, neither of which are expected to recur. Given that the most recent share price is \$65.00, what is a useful AS's trailing price to earnings (P/E) for valuation purposes?

A) 44.52.



B) 37.14.



C) 31.86.



Explanation

Using an underlying earnings concept, an analyst would add back the temporary charges against earnings: $\$1.75 + \$0.19 + \$0.10 = \2.04 . The resulting trailing P/E = $65.00 / 2.04 = 31.86$.

(Module 22.4, LOS 22.e)

Lucas Davenport, CFA, has been assigned the task of doing a valuation analysis of Sanford Systems Inc. Sanford is currently trading at \$15 per share. Exhibit 1 and Exhibit 2 present a summary of Sanford's financial statements for 2007 and 2008.

Davenport has previously completed a FCFE valuation, which yielded a value of \$11.18 per share based on FCFE per common share in 2008 of \$0.85.

Exhibit 1: Sanford Systems Balance Sheets as of 12/31/2008 (in US\$ millions)

	2007	2008
Cash and equivalents	\$325	450
Accounts receivable	850	870
Inventory	<u>1,000</u>	<u>1,050</u>
Total current assets	\$2,175	\$2,370
Gross fixed assets	13,600	15,900
Accumulated depreciation	<u>2,300</u>	<u>2,900</u>
Net fixed assets	11,300	13,000
Total assets	<u>\$13,475</u>	<u>\$15,370</u>
Accounts payable	\$1,500	\$1,520
Notes payable	300	550
Accrued taxes and expenses		
Total current liabilities	\$1,800	\$2,070
Long-term debt	\$5,575	\$6,111

Common stock	100	100
Additional paid-in capital		
Retained earnings	<u>6,000</u>	<u>7,089</u>
Total shareholders' equity	<u>\$6,100</u>	<u>\$7,189</u>
Total liabilities and shareholders' equity	<u>\$13,475</u>	<u>\$15,370</u>

Exhibit 2: Sanford Systems Income Statements for 2007 and 2008 (in US\$ millions)

	2007	2008
Total revenues	\$12,000	\$13,100
Operating costs and expenses	<u>9,400</u>	<u>9,600</u>
EBITDA	\$2,600	\$3,500
Depreciation and amortization	<u>500</u>	<u>600</u>
EBIT	\$2,100	\$2,900
Interest expense	<u>500</u>	<u>585</u>
Income before taxes	\$1,600	\$2,315
Taxes (40%)	<u>640</u>	<u>926</u>
Net income	<u>\$960</u>	<u>\$1,389</u>
Dividends	\$280	\$300
Change in retained earnings	\$680	\$1,089
EPS	\$1.92	\$2.78
DPS	\$0.56	\$0.60
# of shares outstanding (millions)	500	500

Davenport determines that the company follows IFRS rules, and compiles the following industry price-to-adjusted (per share) CFO data, where adjusted CFO is equal to cash flow from operations from the statement of cash flows plus after-tax cash interest expense.

Exhibit 3: Industry Data

	Trailing P/Adjusted CFO per share	Beta	Consensus 5-Year Earnings Growth
Industry Median	2.0x	1.20	9.9%
Sanford		1.25	9.2%

Davenport would also like to make international price multiple comparisons and is contemplating using one or more of the following ratios: price-to-sales, price-to-earnings, price-to-book, price-to-adjusted cash flow

from operations, and enterprise value-to-EBITDA.

Davenport decides to use a single-stage residual income model to estimate the value of Sanford, in addition to the FCFE framework he used earlier. He estimates Sanford's long-term perpetual growth rate in residual income at 5 percent, its return on equity to be 20 percent going forward, weighted average cost of capital to be 10.4 percent based on the target debt-to-asset ratio, and the required return on equity to be 14 percent.

Finally, Davenport solves the following equation for T, given the other inputs (where the index is the S&P 500), and determines that T = 3.6.

$$\ln \left(\frac{\text{Sanford P/E}}{\text{Index P/E}} \right) = T \times \ln \left(\frac{1 + \text{Sanford short-term growth rate} + \text{Sanford dividend yield}}{1 + \text{Index growth rate} + \text{Index dividend yield}} \right)$$

Question #68 - 71 of 140

Question ID: 1586165

Sanford's economic value added (EVA[®]) for 2008 is *closest* to:

- A) \$567.80.
- B) \$1,383.20.
- C) \$525.80.



Explanation

EVA is equal to net operating profit after tax (NOPAT) minus the dollar weighted average cost of capital (\$WACC).

$$\text{NOPAT} = \text{EBIT}(1 - t) = \$2,900(1 - 0.4) = \$1,740$$

$$\text{Invested capital} = \text{LTD} + \text{SH equity} = \$5,575 + \$6,100 = \$11,675$$

$$\text{\$WACC} = \$11,675 \times 0.104 = \$1,214.20$$

$$\text{EVA} = \$1,740 - \$1,214.20 = \$525.80$$

(Module 22.1, LOS 22.a)

Question #69 - 71 of 140

Question ID: 1586166

Based on a comparison of the actual trailing P/FCFE ratio compared to the justified trailing P/FCFE ratio (based on Davenport's FCFE valuation model) for 2008, Sanford is:

- A) overvalued because the actual P/FCFE ratio is greater than the justified P/FCFE ratio for 2008.
- B) correctly valued because the actual P/FCFE ratio is equal to the justified P/FCFE ratio for 2008.
- C) undervalued because the actual P/FCFE ratio is less than the justified P/FCFE ratio for 2008.



Explanation

Sanford's actual P/FCFE ratio is the current market price of \$15 divided by FCFE for 2008:

$$P/FCFE = \frac{\$15.00}{\$0.85} = 17.6x$$

The justified P/FCFE ratio is the value derived from the FCFE valuation model (\$11.18) divided by FCFE for 2008:

$$\text{justified P/FCFE} = \frac{\$11.18}{\$0.85} = 13.1x$$

Based on this analysis, Sanford is overvalued on an absolute basis (NOT relative to the industry benchmark) because the actual P/FCFE ratio is greater than the justified P/FCFE ratio.

(Module 22.1, LOS 22.a)

Question #70 - 71 of 140

Question ID: 1586167

Based on a comparison of the actual trailing P/adjusted CFO ratio compared to the industry median trailing P/adjusted CFO per share ratio for 2008, Sanford:

- is correctly valued relative to the industry benchmark because Sanford's
- A)** P/adjusted CFO ratio is equal to the industry median, despite slightly higher systematic risk and lower 5-year earnings growth. 
- is overvalued relative to the industry benchmark because Sanford's P/adjusted
- B)** CFO ratio is higher than the industry median, despite slightly higher systematic risk and lower 5-year earnings growth. 
- may be undervalued relative to the industry benchmark because Sanford's
- C)** P/adjusted CFO ratio is higher than the industry median, despite slightly higher systematic risk and lower 5-year earnings growth. 

Explanation

Sanford's adjusted CFO is equal to net income plus depreciation minus the increase in net working capital (excluding cash and notes payable) plus after-tax interest expense:

$$\text{adjusted CFO} = \$1,389 + \$600 - \$50 + \$585(1 - 0.4) = \$2,290$$

$$\text{adjusted CFO/share} = \frac{\$2,290}{500} = \$4.58$$

$$P/\text{adjusted CFO} = \frac{\$15}{\$4.58} = 3.3x$$

Sanford is overvalued relative to the industry benchmark because its P/adjusted CFO ratio is higher than the industry median of 2.0, despite slightly higher systematic risk (as measured by beta) and a lower 5-year earnings growth forecast.

(Module 22.1, LOS 22.a)

Question #71 - 71 of 140

Question ID: 1586168

For purposes of this question only, assume Sanford's ROE is 20%, its current market price is \$25, and the cost of equity is 14%. Sanford's implied growth rate in residual income is *closest* to:

- A) 5.11%.
- B) 5.23%.
- C) 5.88%.



Explanation

$$\text{BVPS} = 7,189 / 500 = \$14.38$$

The implied growth rate can be calculated as:

$$g = r - \left[\frac{B_0 \times (\text{ROE} - r)}{V_0 - B_0} \right]$$
$$g = 0.14 - \left[\frac{14.38 \times (0.20 - 0.14)}{25 - 14.38} \right]$$
$$g = 5.88\%$$

(Module 22.1, LOS 22.a)

Question #72 of 140

Question ID: 1473175

The definition of a PEG ratio is price to earnings (P/E):

- A) divided by the average growth rate of the peer group.
- B) divided by the expected earnings growth rate.
- C) divided by average historical earnings growth rate.



Explanation

The PEG ratio is P/E divided by the expected earnings growth rate.

(Module 22.4, LOS 22.j)

Question #73 of 140

Question ID: 1473141

A common justification for using earnings yields in valuation is that:

- A) earnings are more stable than dividends.
- B) negative earnings render P/E ratios meaningless and prices are never negative.
- C) earnings are usually greater than free cash flows.



Explanation

Negative earnings render P/E ratios meaningless. In such cases, it is common to use normalized earnings per share (EPS) and/or restate the ratio as the earnings yield or E/P because price is never negative. Price to earnings (P/E) ranking can then proceed as usual.

(Module 22.4, LOS 22.f)

Question #74 of 140

Question ID: 1586176

Robert Chan comments to Leslie Singer that Converted Industries' expected dividend growth rate is 5.0%, dividend payout ratio (g) is 45%, and required return on equity (r) is 10%. Based on a justified trailing P/E ratio compared to the stock's trailing P/E ratio at market of 9.0, Converted Industries is *most likely*:

A) undervalued.



B) correctly valued.



C) overvalued.



Explanation

Justified trailing P/E = payout ratio * $(1 + g) / (r - g)$. When the expected dividend growth is 5.0%, the justified trailing P/E = $0.45 * (1 + 0.05) / (0.10 - 0.05) = 9.45$. This is greater than the market P/E of 9.0.

(Module 22.2, LOS 22.I)

Question #75 of 140

Question ID: 1473164

A firm's return on equity (ROE) is 14%, its required rate of return is 10%, and its expected growth rate is 8%. What is the firm's justified price-to-book value (P/B) based on these fundamentals?

A) 3.00.



B) 2.00.



C) 2.75.



Explanation

The firm's justified price-to-book value = $(ROE - g) / (r - g) = (0.14 - 0.08) / (0.10 - 0.08) = 3.00$

(Module 22.4, LOS 22.i)

Question #76 of 140

Question ID: 1473192

Proprietary Technologies, Inc., (PTI) has a leading price-to-earnings (P/E) ratio of 28 while the median leading P/E of a peer group of companies within the industry is 28. Based on the method of comparables, an analyst would *most likely* conclude that PTI should be:

- A) bought as an undervalued stock. 
- B) sold or sold short as an overvalued stock. 
- C) viewed as a properly valued stock. 

Explanation

The price per dollar of earnings is the same as that for the median of the peer group, which implies that it is likely properly valued.

(Module 22.4, LOS 22.m)

Question #77 of 140

Question ID: 1473147

An analyst is valuing a company with a dividend payout ratio of 0.65, a beta of 0.72, and an expected earnings growth rate of 0.05. A regression on comparable companies produces the following equation:

Predicted price to earnings (P/E) = $7.65 + (3.75 \times \text{dividend payout}) + (15.35 \times \text{growth}) - (0.70 \times \text{beta})$

What is the predicted P/E using the above regression?

- A) 7.65. 
- B) 10.35. 
- C) 11.39. 

Explanation

Predicted P/E = $7.65 + (3.75 \times 0.65) + (15.35 \times 0.05) - (0.70 \times 0.72) = 10.35$

(Module 22.4, LOS 22.h)

Question #78 of 140

Question ID: 1473094

P/E multiples are often computed using the average of the multiples of comparable firms, because:

- A) it is very easy to find comparable firms that have the same business mix and risk and growth profiles. 
- B) it is conceptually very straightforward. 
- C) it provides the most accurate results. 

Explanation

The use of comparable firms is quite common, because it is conceptually very straightforward. Also, it does not require the analyst to make specific assumptions regarding growth, risk, and other variables. However, it is often difficult to find comparable firms, since even within the same industry different firms can have different business mixes and risk and growth profiles.

(Module 22.1, LOS 22.a)

Question #79 of 140

Question ID: 1473109

An increase in return on equity (ROE) will cause a price-to-book (P/B) multiple to:

A) decrease.



B) there is insufficient information to tell.



C) increase.

**Explanation**

An increase in ROE should increase the price to book (P/B) ratio:

$$P_0 / B_0 = (ROE - g) / (r - g)$$

(Module 22.2, LOS 22.g)

Question #80 of 140

Question ID: 1586173

What is the justified trailing price-to-earnings (P/E) multiple of a stock that has a payout ratio of 65% if the shareholders require a return of 10% on their investment and the expected growth rate in dividends is 6%?

A) 9.28.



B) 17.23.



C) 16.25.

**Explanation**

$$P_0/E_0 = (0.65 \times 1.06) / (0.10 - 0.06) = 17.225$$

(Module 22.2, LOS 22.i)

Question #81 of 140

Question ID: 1473121

A decrease in the earnings retention rate will cause a price-to-sales (P/S) multiple to:

A) increase.



B) decrease.



C) remain the same.

**Explanation**

A decrease in the earnings retention rate will increase the following expression for P/S due to the implied increase in the payout ratio, which is $(1 - b)$:

$$P_0 / S_0 = [(E_0 / S_0) (1 - b)(1 + g)] / (r - g)$$

Note that the topic review does not allow for any interactive relationship between retention and growth. Thus, no explicit consideration is given to whether the increase in the payout ratio will cause an offsetting decrease in growth.

(Module 22.3, LOS 22.g)

Question #82 of 140

Question ID: 1473210

An analyst gathered the following data for TRK Construction [all amounts in Swiss francs (Sf)]:

Recent share price	Sf 25.00
Shares outstanding	40 million
Market value of debt	Sf 130 million
Cash and marketable securities	Sf 65 million
Investments	Sf 250 million
Net income	Sf 150 million
Interest expense	Sf 8 million
Depreciation and amortization	Sf 11 million
Taxes	Sf 52 million

The EV/EBITDA multiple for TRK Construction is *closest* to:

A) 2.47x.



B) 3.69x.



C) 4.12x.



Explanation

EBITDA = (net income + interest + taxes + depreciation / amortization)

EV = (market value of common stock + market value of debt – cash and investments)

EBITDA = 150 + 8 + 11 + 52 = Sf 221 million

EV = $(25 \times 40) + 130 - 65 - 250$ = Sf 815 million

EV / EBITDA = 3.69

(Module 22.4, LOS 22.o)

Question #83 of 140

Question ID: 1473212

Which of the following price multiples is *most* severely damaged by international accounting differences?

- A) Price to free cash flow to equity (P/FCFE). 
- B) Price to cash flow from operations (P/CFO). 
- C) Enterprise value to earnings before interest, taxes, depreciation, and amortization (EV/EBITDA). 

Explanation

EV/EBITDA is the most seriously affect because it is most closely tied to accounting conventions.

(Module 22.4, LOS 22.p)

Question #84 of 140

Question ID: 1473140

A common pitfall in interpreting earnings yields in valuation is:

- A) using negative earnings. 
- B) using underlying earnings. 
- C) look-ahead bias. 

Explanation

A common pitfall is look-ahead bias, wherein the analyst uses information that was not available to the investor when calculating the earnings yield.

(Module 22.4, LOS 22.f)

Question #85 of 140

Question ID: 1473196

Enhanced Systems, Inc., (ESI) has a price to book value (P/B) of four while the median P/B of the stock market overall is three, and the median P/B of companies within the industry is six. Based on the method of comparables, an analyst would *most likely* conclude that ESI:

- A) should be purchased because it is an undervalued stock. 
- B) is of indeterminate relative value, due to conflicting metrics. 
- C) should be sold because it is an overvalued stock. 

Explanation

The price per dollar of book value of ESI is considerably lower than that for the median of the peer group, which implies that it may well be undervalued. For the method of comparables, we most appropriately select as comparison assets companies operating in the same industry as the subject company.

(Module 22.4, LOS 22.m)

Question #86 of 140

Question ID: 1473090

An analyst begins an equity analysis of Company A by noting the following ratios from three companies in the same industry:

	EPS	PE
Company A	\$1.60	10.0
Company B	\$2.10	12.5
Company C	\$5.80	13.0

This analyst is *most likely* using:

- A) the method of forecasted fundamentals.
- B) the method of comparables.
- C) technical analysis.

**Explanation**

The analysis is comparing ratios of three companies in the same industry. The Law of One Price states that similar assets should have comparable prices.

(Module 22.1, LOS 22.a)

Question #87 of 140

Question ID: 1473093

Which of the following statements about the method of comparables in price multiple valuation is CORRECT?

- A) It assumes that cash flows are related to fundamentals.
- B) It values an asset relative to a benchmark value of the multiple.
- C) It relates multiples to company fundamentals using a discounted cash flow (DCF) model.

**Explanation**

The method of comparables involves using a price multiple to evaluate whether an asset is valued properly *relative* to a benchmark value of the multiple. It makes no explicit assumptions about fundamentals and does not rely on a DCF model.

(Module 22.1, LOS 22.a)

Question #88 of 140

Question ID: 1473185

A common price to earnings (P/E) based method for estimating terminal value in multi-stage models is the:

- A) fundamentals approach.



B) dividend yield approach.



C) P/E to growth (PEG) approach.



Explanation

It is common to restate the Gordon growth model price as a multiple of expected future book value per share or earnings per share (EPS).

(Module 22.4, LOS 22.k)

Question #89 of 140

Question ID: 1473203

If cash flow from operations (CFO) embeds financing-related flows, it should be adjusted by:

A) subtracting capital expenditures.



B) subtracting (net interest outflow) $\times$ (1 - tax rate).



C) adding (net interest outflow) $\times$ (1 - tax rate).



Explanation

Cash flow from operations CFO should be adjusted to $\text{CFO} + (\text{net cash interest outflow}) \times (1 - \text{tax rate})$, if CFO embeds financing-related flows.

(Module 22.4, LOS 22.n)

Question #90 of 140

Question ID: 1473215

Which of the following is a common momentum valuation indicator?

A) Dividend yield (D/P).



B) Price to free cash flow to equity (P/FCFE).



C) Relative strength.



Explanation

Relative strength is generally considered a momentum valuation indicator.

(Module 22.4, LOS 22.q)

Question #91 of 140

Question ID: 1473115

Analyst Ariel Cunningham likes using the price/earnings ratio for valuation purposes because studies have shown it is very effective at identifying undervalued stocks. However, she has one main problem with the statistic – it doesn't work when a company loses money. So Cunningham is considering switching to a different core valuation metric. Given Cunningham's rationale for using the price/earnings ratio, which option would be her *best* alternative?

A) Price/sales.



B) Price/cash flow.



C) Price/book.



Explanation

Book value is usually positive, but not always. Cash flow is often negative. If the reason Cunningham wants to stop using the P/E ratio is that it does not work for unprofitable companies, her best option is a ratio base on sales, which are positive in all but the rarest of instances.

(Module 22.3, LOS 22.c)

Question #92 of 140

Question ID: 1473098

The warranted or intrinsic price multiple is called the:

A) multiple implied by historical growth.



B) multiple implied by the market price.



C) justified price multiple.



Explanation

A justified price multiple is the warranted or intrinsic price multiple. It is the estimated fair value of that multiple.

(Module 22.1, LOS 22.b)

Question #93 of 140

Question ID: 1586174

What is the justified trailing price-to-earnings (P/E) multiple of a stock that has a payout ratio of 40% if the shareholders require a return of 16% on their investment and the expected growth rate in dividends is 6%?

A) 4.00.



B) 4.24.



C) 6.36.



Explanation

$$P_0/E_0 = (0.40 \times 1.06) / (0.16 - 0.06) = 4.24$$

(Module 22.2, LOS 22.i)

Question #94 of 140

Question ID: 1473211

Which of the following factors is NOT a source of differences in cross-border valuation comparisons?

- A) Cultures. 
- B) Growth opportunities. 
- C) Intra-country market indicators. 

Explanation

Intra-country market indicators are not, by definition, cross-border.

(Module 22.4, LOS 22.p)

Question #95 of 140

Question ID: 1473104

The trailing price-to-earnings (P/E) ratio is defined as:

- A) price to most recent earnings. 
- B) price to next period's expected earnings. 
- C) the average P/E over the last five years. 

Explanation

The trailing P/E ratio is price to most recent realized earnings.

(Module 22.1, LOS 22.d)

Question #96 of 140

Question ID: 1473195

Enhanced Systems, Inc., (ESI) has a leading price to sales (P/S) of 0.18 while the median leading P/S of a peer group of companies within the industry is 0.10. Based on the method of comparables, an analyst would *most likely* conclude that ESI should be:

- A) sold or sold short as an overvalued stock. 
- B) bought as an undervalued stock. 
- C) bought on margin as an undervalued stock. 

Explanation

The price per dollar of sales is considerably higher than that for the median of the peer group, which implies that it may well be overvalued.

(Module 22.4, LOS 22.m)

Question #97 of 140

Question ID: 1473176

The relative valuation model known as the PEG ratio is equal to:

- A) price-to-earnings (P/E) / earnings per share (EPS) growth rate. 
- B) earnings per share growth rate / price-to-earnings. 
- C) $P/E \times \text{earnings}$. 

Explanation

The PEG ratio is equal to the price-to-earnings ratio divided by the EPS growth rate.

(Module 22.4, LOS 22.j)

Question #98 of 140

Question ID: 1473100

An argument for using the price-to-earnings (P/E) valuation approach is that:

- A) management discretion increases the reliability of the ratio. 
- B) earnings power is the primary determinant of investment value. 
- C) earnings can be negative. 

Explanation

Earnings power is the primary determinant of investment value. Both remaining factors reduce the usefulness of the P/E approach.

(Module 22.1, LOS 22.c)

Question #99 of 140

Question ID: 1473191

Proprietary Technologies, Inc., (PTI) has a leading price-to-earnings (P/E) ratio of 38 while the median leading P/E of a peer group of companies within the industry is 28. Based on the method of comparables, an analyst would *most likely* conclude that PTI should be:

- A) viewed as a properly valued stock. 
- B) bought as an undervalued stock. 
- C) sold or sold short as an overvalued stock. 

Explanation

The price per dollar of earnings is considerably higher than that for the median of the peer group, which implies that it may well be overvalued.

(Module 22.4, LOS 22.m)

Question #100 of 140

Question ID: 1473143

An increase in profit margin will cause a price-to-sales (P/S) multiple to increase if:

- A) there is insufficient information to tell. 

B) the growth rate in sales does not decrease proportionately.



C) the required rate of return increases.



Explanation

An increase (decrease) in the profit margin increases (decreases) the growth rate if sales do not decrease (increase) proportionately. Increases in the required rate of return would decrease the P/S ratio. This is clear in the expression for trailing P/S:

$$P_0 / S_0 = [(E_0 / S_0)(1 - b)(1 + g)] / (r - g)$$

(Module 22.4, LOS 22.g)

Question #101 of 140

Question ID: 1473177

Good Sports, Inc., (GSI) has a leading price-to-earnings (P/E) ratio of 12.75 and a 5-year consensus growth rate forecast of 8.5%. What is the firm's P/E to growth (PEG) ratio?

A) 1.50.



B) 0.67.



C) 150.00.



Explanation

The firm's PEG is $12.75 / 8.50 = 1.50$.

(Module 22.4, LOS 22.j)

Question #102 of 140

Question ID: 1473174

Two security analysts, Ramon Long and Sri Beujeau, disagree about certain aspects of the PEG ratio. Long argues that: "unlike typical valuation metrics that incorporate dividend discounting, the PEG ratio is unique because it generates meaningful results for firms with negative expected earnings-growth." Is Long correct?

A) Yes, because the expected earnings-growth rate is cancelled out in the computation of the PEG ratio.



B) No, because the PEG ratio generates meaningless results for negative earnings-growth companies.



C) Yes, because the computation of the PEG ratio does not use the rate of expected earnings growth.



Explanation

The PEG ratio is: $PEG = (P/E) / \text{earnings growth}$. As such, firms with negative expected earnings growth will have a negative PEG ratio, which is meaningless.

(Module 22.4, LOS 22.j)

The Farmer Co. has a payout ratio of 65% and a return on equity (ROE) of 16% (assume that this is expected ROE for the upcoming year). What will be the appropriate price-to-book value (PBV) based on return differential if the expected growth rate in dividends is 5.6% and the required rate of return is 13%?

A) 1.41.



B) 0.71.



C) 1.48.



Explanation

Based on return differential:

$$P_0 / BV_0 = (ROE_1 - g) / (r - g) = (0.16 - 0.056) / (0.13 - 0.056) = 1.41.$$

(Module 22.4, LOS 22.i)

Analysts and portfolio managers at Big Picture Investments are having their weekly investment meeting. CEO Bob Powell, CFA, believes the firm's portfolios are too heavily weighted toward growth stocks. "I expect value to make a comeback over the next 12 months. We need to get more value stocks in the Big Picture portfolios." Four of Powell's analysts, all of whom hold the CFA charter, were at the meeting – Laura Barnes, Chester Lincoln, Zelda Marks, and Thaddeus Bosley. Powell suggested Big Picture should start selecting stocks with the lowest price-to-earnings (P/E) multiples. Here are the analysts' comments:

- Barnes said numerous academic studies have shown that low P/E stocks tend to outperform those with high P/Es. She uses the P/E ratio as the basis of most of her valuation analysis.
- Lincoln warned against using P/E ratios to evaluate technology stocks. He suggests using price-to-book (P/B) ratios instead, because they are useful for explaining long-term stock returns.
- Bosley prefers the price/sales (P/S) ratio and the earnings yield.
- Marks acknowledges that the P/E ratio is a useful valuation measurement. However, she prefers using the price/free-cash-flow ratio.

Powell has provided Barnes with a group of small-cap stocks to analyze. The stocks come from a variety of different sectors and have widely different financial structures and growth profiles. She has been asked to determine which of these stocks represent attractive values. She is considering four possible methods for the job:

- The PEG ratio, because it corrects for risk if the stocks have similar expected returns.
- Comparing P/E ratios to the average stock in the Russell 2000 Index, because the benchmark should serve as a good proxy for the average small-cap stock valuation.
- Comparing P/E ratios to the median stock in the Russell 2000 Index, because outliers can skew the average P/E upward.
- The P/S ratio, because it works well for companies in different stages of the business cycle.

Barnes is contemplating the use of a price/earnings ratio to value a start-up medical technology firm. Which of the following is the *most* compelling reason not to use the P/E ratio?

- A) P/E ratios for medical-technology firms with different specialties are not comparable. 
- B) The company is likely to be unprofitable. 
- C) Earnings per share are not a good determinant of investment value for medical-technology companies. 

Explanation

Earnings are the chief determinant of value for most companies, including med-tech. P/E is the most common valuation method and the best known by lay investors. Comparability of P/E ratios across industries is always problematic, but not as much so for within the med-tech industry. A start-up company is very likely to have negative earnings, which renders the P/E ratio useless.

(Module 22.4, LOS 22.c)

Question #105 - 107 of 140

Question ID: 1473128

Based on their responses to Powell, which of the analysts is *most likely* concerned about earnings volatility?

- A) Lincoln. 
- B) Barnes. 
- C) Bosley. 

Explanation

Book value tends to be more stable than earnings. Therefore, Lincoln's favorite valuation tool, the P/B ratio, is less volatile than the P/E. The P/S ratio tends to be less volatile than the P/E as well, but Bosley's other favorite, earnings yield, is just as volatile. The method preferred by Barnes is likely to be more volatile than the P/B ratio.

(Module 22.4, LOS 22.c)

Question #106 - 107 of 140

Question ID: 1473129

Barnes would be *least likely* to use EV/EBITDA ratio, rather than the P/E ratio, when analyzing a company that:

- A) pays a dividend, and is likely to deliver little earnings growth. 
- B) reports a lot of depreciation expense. 
- C) has a different capital structure than most of its peers. 

Explanation

For companies that report a lot of depreciation expense or must be compared to companies with different levels of financial leverage, the EV/EBITDA ratio may be more useful than the P/E. For companies that pay a dividend and have little profit growth, both should work fine. Given Barnes' stated preference for the P/E ratio, she is least likely to use the EV/EBITDA ratio with the dividend-paying firm.

(Module 22.4, LOS 22.c)

Question #107 - 107 of 140

Question ID: 1473130

Barnes is considering the four methods previously described to analyze the small-cap stocks provided to her by Powell. For which method does Barnes provide the *weakest* justification?

- A) The price/sales ratio. 
- B) The mean P/E of Russell 2000 companies. 
- C) The PEG ratio. 

Explanation

No valuation method will work dependably across all types of stocks. The four Barnes proposed are probably as good as any. But the PEG ratio does not correct for risk – it works as a comparison tool only if the companies have similar expected risks and returns. The other justifications are reasonable.

(Module 22.4, LOS 22.c)

Question #108 of 140

Question ID: 1473107

Which of the following is a *disadvantage* of using the price-to-book value (PBV) ratio?

- A) Book values are affected by accounting standards, which may vary across firms and countries. 
- B) Firms with negative earnings cannot be evaluated with the PBV ratios. 
- C) Book value may not mean much for manufacturing firms with significant fixed costs. 

Explanation

The disadvantages of using PBV ratios are:

1. Book values are affected by accounting standards, which may vary across firms and countries.
2. Book value may not mean much for service firms without significant fixed costs.
3. Book value of equity can be made negative by a series of negative earnings, which limits the usefulness of the variable.

(Module 22.2, LOS 22.c)

Question #109 of 140

Question ID: 1473096

Which of the following statements about the method of forecasted fundamentals in price multiple valuation is *most* accurate?

- A) It relies on the Law of One Price. 
- B) It values an asset relative to a benchmark value of the multiple. 
- C) It relates multiples to company fundamentals using a discounted cash flow (DCF) model. 

Explanation

The method of forecasted fundamentals relates multiples to company fundamentals using a DCF method. It does not explicitly rely on the Law of One Price. Further, it does not typically focus on benchmarks.

(Module 22.1, LOS 22.a)

Question #110 of 140

Question ID: 1473105

At a CFA society function, Andrew Caza comments to Nanda Dhople that the expected dividend growth rate (g) for Zeron Enterprises Inc (ZEI) is expected increase 0.5% from 6% to 6.5%. Caza claims that since ZEI will maintain their historic dividend payout ratio (g) of 50% and cost of equity (k) of 10%, ZEI's P/E ratio will also increase by 0.5%. Is Caza *correct*?

- A) No, ZEI's P/E ratio will increase by approximately 14.32%. 
- B) Yes, ZEI's P/E ratio will increase by approximately 0.5%. 
- C) No, ZEI's P/E ratio will decrease by approximately 14.32%. 

Explanation

Caza is not correct. $P/E_{ZEI} = \text{payout ratio} / (k - g)$

When the expected dividend growth is 6%, $P/E = 0.50 / (0.10 - 0.06) = 12.50$

When the expected dividend growth is 6.5%, $P/E = 0.50 / (0.10 - 0.065) = 14.29$

The percentage change is $(14.29 / 12.50) - 1 = 14.32\%$, representing a 14.32% increase.

(Module 22.1, LOS 22.d)

Question #111 of 140

Question ID: 1473113

An analyst has gathered the following data about the Garber Company:

- Payout Ratio = 60%.
- Expected Return on Equity = 16.75%.
- Required rate of return = 12.5%.

What will be the appropriate price-to-book value (PBV) ratio for the Garber Company based on return differential?

A) 1.73.



B) 0.58.



C) 1.38.



Explanation

The estimated growth rate is 6.7% [$0.1675 \times (1 - 0.60)$] and PBV ratio based on rate differential will be:

$$P_0 / BV_0 = (ROE_1 - g) / (r - g) = (0.1675 - 0.067) / (0.125 - 0.067) = 1.73.$$

(Module 22.2, LOS 22.i)

Question #112 of 140

Question ID: 1473161

What is the appropriate price-to-sales (P/S) multiple of a stock that has a retention ratio of 45%, a return on equity (ROE) of 14%, an earnings per share (EPS) of \$5.25, sales per share of \$245.54, an expected growth rate in dividends and earnings of 6.5%, and shareholders require a return of 11% on their investment?

A) 0.227.



B) 0.158.



C) 0.278.



Explanation

Recall that profit margin is measured as E_0 / S_0 . In this example, the profit margin is $(5.25 / 245.54) = 0.0214$. Thus:

$$P_0 / S_0 = [(E_0 / S_0)(1 - b)(1 + g)] / (r - g) = [0.0214(0.55)(1.065)] / (0.11 - 0.065) = 0.278$$

(Module 22.4, LOS 22.i)

Question #113 of 140

Question ID: 1473219

An analyst is calculating the weighted harmonic mean P/E ratio of a 2-stock portfolio. Stocks AAA and BBB have prices of \$12 and \$15, respectively, and EPS of \$1 and \$2, respectively. Which of the following is the weighted harmonic mean P/E of the portfolio *closest* to?

A) 9



B) 9.75



C) 9.23



Explanation

The weighted harmonic mean is $1/[(12/27)(1/12) + (15/27)(2/15)] = 27/3 = 9.00$. The weighted harmonic mean of the individual stocks P/Es is the best measure of the P/E for a portfolio of stocks.

(Module 22.4, LOS 22.r)

At the end of 2x09, Dustin Pedroia, CFA, is writing a report to help advise on a potential corporate takeover. Iliot Inc. is up for sale, and Pedroia's client is considering buying 100% of the share capital.

Pedroia has decided to include two free cash flow valuations in his report for the client. An extract of the most recent cash flow statement, which he intends to use as a base for his first FCF calculation, appears below:

Cash Flow Statement (extract) for the Year Ended 31st December 2x09

U.S. \$ millions

Cash flow from operating activities 130

Extracts from the Financial Statements for 2x09 also show the following:

Financial Statement Extracts	2x08	2x09
	\$m	\$m

Fixed assets (at cost) 270.0 320.0

Less: Accumulated depreciation 112.0 138.0

Inventory 65.2 71.0

Accounts receivable 94.2 96.7

Accounts payable 74.0 79.0

There have been no sales or impairments of fixed assets during the year and net borrowing for 2x09 raised \$14 million.

For his first valuation, Pedroia will make a simple assumption that free cash flow to equity will grow at 5% per annum indefinitely in order to reach his valuation. The resulting value will be labelled "Best Case Scenario" in the report.

In addition, the client has passed Pedroia their own forecasts for the performance of Iliot over the next five years, and he also intends to use these forecasts to come up with an alternative valuation, which he will label "FCF Valuation Using Forecasted Cash Flows." Details of the forecasted flows are as follows:

Free Cash Flow to Equity Forecasts: Iliot Inc.

2x10 \$65 million

2x11 \$68 million

2x12 \$72 million

2x13 \$75 million

Pedroia will discount the flows at a cost of equity of 12%, and that the 2x10 free cash flow will occur in one year from now. In order to calculate a terminal value at the end of 2x13, Pedroia intends to use an estimate of Iliot's P/E ratio and earnings. He estimates Iliot's trailing P/E ratio at the end of 2x13 to be 28 using a linear regression model based on risk, growth, and dividend payout, and forecasts 2x13 earnings to be \$70 million.

Pedroia also wishes to include a note on Iliot's normalized earnings in his final report. He intends to initially calculate a normalized EPS figure for 2x09. To do this he will use the method of average return on equity method. In order to assist with this task he notes down various information for Iliot from the last three years:

Iliot Historical Data	2x07	2x08	2x09
Earnings per share	\$2.80	\$2.50	\$2.85
Book value per share	\$16.20	\$15.80	\$16.40
Return on equity	14.8%	15.4%	18.0%

Pedroia intends to conclude his report with a note to the client that he himself owns a small number of Iliot shares. He purchased the shares after implementing a stock screen system of selection, whereby he decided to only purchase shares if they passed the following criteria:

- P/E less than 10
- Market Cap greater than \$0.5 billion
- EBITDA-to-free cash flow ratio less than 12
- PEG ratio greater than 1.2

He implemented his stock screen system in mid 2x06. Before implementing the system, Pedroia back tested it using 2x05 year-end ratios published by his favorite analyst's journal in April 2x06. Using those ratios, results showed that if he had bought stocks at the end of 2x05, which passed his screen, he would have made abnormal positive profits.

Calculate free cash flow to equity during 2x09 using the data extracted from the 2x09 accounts:

- A) \$90.7 million.
- B) \$94.0 million.
- C) \$120.0 million.



Explanation

FCFE = CFO – FC Inv + net borrowing

= 130 – 50 + 14

= \$94 million

(Module 21.4, LOS 21.d)

Question #115 - 119 of 140

Question ID: 1473222

The growth assumption Pedroia uses in calculating his "Best Case Scenario" valuation are most suitable if Iliot is:

- A) a stable firm in a mature industry with a required return on equity of 4%.
- B) a stable firm in a mature industry with a required return on equity of 14%.
- C) a growing firm in an infant industry with a required return on equity of 14%.



Explanation

The assumption of constant perpetual growth is suited to stable firms in a mature industry. If Iliot has a cost of equity of 4%, this would be less than the growth rate (5%) assumed and hence the model would not be appropriate.

(Module 20.3, LOS 20.I)

Question #116 - 119 of 140

Question ID: 1473223

Calculate the value of equity to the nearest \$1 million using a FCFE model and the cash flows / assumptions that Pedroia uses in his "FCF Valuation Using Forecasted Cash Flows" valuation.

- A) \$1,457 million.
- B) \$2,171 million.
- C) \$1,620 million.



Explanation

PV of initial cash flows discounted at 12%:

CF0 nil

C01 65

C02 68

C03 72

C04 75

I = 12%

PV = 211.16

Terminal value at 2013 = $28 \times 70 = 1,960$

$PV = 1,960 / (1.12)^4 = 1,245.62$

Value = $1,245.62 + 211.16 = \mathbf{1,456.78}$

(Module 21.5, LOS 21.I)

Question #117 - 119 of 140

Question ID: 1473224

Which of the following is the normalized earnings figure for 2x10, which will be calculated by Pedroia using the average return on equity method?

A) \$2.63.



B) \$2.72.



C) \$2.59.



Explanation

Using the average return on equity method, normalized EPS is calculated as the average return on equity multiplied by the current book value per share.

Average ROE = $(14.8 + 15.4 + 18.0) / 3 = 16.07\%$

Normalized EPS = $0.1607 \times 16.40 = \mathbf{\$2.63}$

(Module 22.4, LOS 22.e)

Question #118 - 119 of 140

Question ID: 1473225

Which of the following is least likely to be a limitation of the predicted P/E used by Pedroia to calculate the terminal value in his "FCF Valuation Using Forecasted Cash Flows" valuation?

- A) The predictive power of the estimated regression for a different time period is uncertain. 
- B) The relationship between P/E and the fundamental variables examined will be static. 
- C) Multicollinearity is often a problem in time series regressions such as the one Pedroia has used. 

Explanation

A common limitation is that the relationships may change over time rather than remain static.

(Module 22.4, LOS 22.h)

Question #119 - 119 of 140

Question ID: 1473226

Which of the following errors has Pedroia made in back testing his stock screen?

- A) His results are subject to survivorship bias. 
- B) His results are likely to suffer from multicollinearity. 
- C) His results are likely to suffer from look ahead bias. 

Explanation

Pedroia should have assumed that he could purchase the stocks on the day the ratios were published, not the year-end. As a result of this mistake, he is exposed to the potential effects of look-ahead bias.

(Module 22.4, LOS 22.r)

Question #120 of 140

Question ID: 1473207

An analyst gathered the following data for TRK Construction [all amounts in Swiss francs (Sf)]:

Recent share price	Sf 22.00
Shares outstanding	40 million
Market value of debt	Sf 140 million
Cash and marketable securities	Sf 55 million
Investments	Sf 300 million
Net income	Sf 140 million
Interest expense	Sf 7 million
Depreciation and amortization	Sf 10 million
Taxes	Sf 56 million

The EV/EBITDA ratio for TRK Construction is *closest* to:

A) 3.12x.



B) 2.52x.



C) 3.49x.



Explanation

EBITDA = (net income + interest + taxes + depreciation / amortization)

EV = (market value of common stock + market value of debt – cash and investments)

EBITDA = 140 + 7 + 10 + 56 = Sf 213 million

EV = (22 × 40) + 140 – 55 – 300 = Sf 665 million

EV / EBITDA = 3.12

(Module 22.4, LOS 22.o)

Question #121 of 140

Question ID: 1473213

Which of the following factors is a source of differences in cross-border valuation comparisons?

A) Comparative advantage.



B) Accounting methods.



C) Intra-country market indicators.



Explanation

Different accounting conventions make cross-border comparisons for valuation purposes challenging.

(Module 22.4, LOS 22.p)

Question #122 of 140

Question ID: 1473101

A firm is better valued using the discounted cash flow approach than the P/E multiples approach when:

A) dividend payout is low.



B) expected growth rate is very high.



C) earnings per share are negative.

**Explanation**

P/E multiples are not meaningful when the earnings per share are negative. While this problem can be partially offset by using normalized or average earnings per share, the problem cannot be eliminated.

(Module 22.1, LOS 22.c)

Question #123 of 140

Question ID: 1473136

Glad Tidings Gifts (GTG) recently reported annual earnings per share (EPS) of \$2.25, which included an extraordinary loss of \$0.17 and an expense of \$0.12 related to acquisition costs during the accounting period, neither of which are expected to recur. Given that the most recent share price is \$50.00, what is a useful GTG's trailing price to earnings (P/E) for valuation purposes?

A) 22.22.



B) 19.69.



C) 25.51.

**Explanation**

Using an underlying earnings concept, an analyst would add back the temporary charges against earnings: $\$2.25 + \$0.17 + \$0.12 = \2.54 . The resulting trailing P/E = $50.00 / 2.54 = 19.69$.

(Module 22.4, LOS 22.e)

Victoria Banks is a senior analyst working for a large firm of portfolio managers. Her manager, David Alan, has asked her to report on a company called Retro Inc. as he believes it might offer a potentially good investment. The accounts for Retro Inc. are given below.

Retro prepares its accounts using U.S. GAAP.

Exhibit 1: Retro Inc. Balance Sheet as at 31 December

	20x9	20x8
	\$m	\$m

Assets

Cash	150	100
Accounts receivable	1,700	1,620
Inventory	<u>1,810</u>	<u>1,800</u>
Total current assets	3,660	3,520
Property, plant, and equipment	1,430	1,000
Intangibles	<u>100</u>	<u>150</u>
Total assets	5,190	4,670
Liabilities and Capital		
Notes payable to banks	200	220
Accounts payable	1,330	1,200
Interest payable	<u>130</u>	<u>100</u>
Total current liabilities	1,660	1,520
Long-term debt	770	680
Deferred tax	820	790
Common stock	1,300	1,300
Retained earnings	<u>640</u>	<u>380</u>
Total liabilities and capital	5,190	4,670

Exhibit 2: Retro Inc. Income Statement for the Year Ended 31 December 20x9

	\$m
--	------------

Sales	3,000
-------	-------

Cost of goods sold	(1,800)
Gross profit	1,200
Depreciation	(150)
Amortization	(50)
SG&A	(280)
Gain on disposal	30
Restructuring charge reversal	20
Interest expense	(190)
Income tax expense	(223)
Net income	357

Retro disposed of PPE in the year that had a cost of \$150m and accumulated depreciation at the time of disposal of \$90m. No intangibles were disposed of during the year. Deferred tax liabilities are not expected to reverse for the foreseeable future.

Banks is also concerned that the net income looks relatively high when compared to previous years and therefore wants to measure the quality of earnings. She has heard that the lower the accruals ratio the higher the quality of earnings.

Banks calculates that Retro Inc. has a leading P/E ratio of 4.29 and a five-year consensus growth rate forecast at 14.85%. The median PEG, based on leading P/E, for a group of companies comparable in risk to Retro Inc. is 0.82. Based on this Banks wants to determine whether the stock is correctly priced.

One of Banks's colleagues, Jennifer Cery, comments that P/E multiples are not always that useful and that sometimes enterprise value multiples are better. She makes the following comments:

Comment 1: Enterprise value multiples are useful when comparing firms with different degrees of financial leverage and when EPS is negative.

Comment 2: As EBITDA can be used as a proxy for free cash flow to the firm providing depreciation is close to capital expenditure and the firms levels of working capital is relatively constant.

Question #124 - 127 of 140

Question ID: 1586189

Calculate free cash flow to equity (FCFE):

A) 37.



B) 127.



C) 57.



Explanation

Calculation of CFO:

$$\text{CFO} = \text{NI} + \text{NCC} - \text{WC}_{\text{INV}}$$

$$\text{CFO} = \$357 + \$180 + \$70 = \$607$$

Depreciation	150		2010	2009
Amortization	50	Current assets	3,660	3,520
Gain on asset disposal	(30)	Cash	<u>(150)</u>	<u>(100)</u>
Reversal of provision	(20)		3,510	3,420
↑DTL	<u>30</u>			
Total non cash charges	180	Current liabilities	1,660	1,520
		Notes payable	<u>(200)</u>	<u>(220)</u>
			<u>1,460</u>	<u>1,300</u>
		Working capital	2,050	2,120
		WC _{INV}	-70	

Note that the change in the DTL liability is only included as a non-cash charge (NCC) as it is not expected to reverse in the foreseeable future. If the DTL is expected to reverse in the short run it should be ignored when adding back NCCs.

Calculation of CFI:

$\text{FC}_{\text{INV}} = \text{change in NBV (net PP\&E)} + \text{depreciation and amortization expense} - \text{gain on disposal}$

$$\text{FC}_{\text{INV}} = \$380 + \$200 - \$30 = \$550\text{m}$$

Alternative using reconciliation approach:

Opening PPE were \$1,000m, these were depreciated by \$150m and the closing PPE were \$1,430. Since the disposal had a NBV of \$60m the company must have spent:

PPE:

NBV 2009 b/fwd	1,000
NBV of disposal	(60)
Depreciation expense	(150)

Balancing figure 'Additions' **640**

NBV 2010 c/fwd	1,430
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On the disposal:

Proceeds (Balancing figure)	90
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<u>NBV of disposal</u>	<u>60</u>
------------------------	-----------

Gain on disposal	30
------------------	----

Intangibles:

NBV 2009 b/fwd	150
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Disposals	(0)
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Amortization expense	(50)
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<u>Balancing figure 'Additions'</u>	<u>0</u>
--	-----------------

NBV 2010 c/fwd	100
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Additions	(640)
-----------	-------

<u>Proceeds on disposal</u>	<u>90</u>
-----------------------------	-----------

CFI	(550)
------------	--------------

Change in debt	70
-----------------------	-----------

Free cash flow for equity

CFO	607
-----	-----

CFI	(550)
-----	-------

Change in debt	<u>70</u>
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FCFE	127
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Question #125 - 127 of 140

Question ID: 1473182

Using the cash flow statement approach calculate the aggregate accruals ratio:

A) 11.5%.



B) 5.7%.



C) 10.2%.

**Explanation**

Aggregate accruals = NI – CFO – CFI

Aggregate accruals = \$357 – \$607 + 550 = \$300

Accruals ratio = \$300 / (\$2,760 + \$2,480) / 2 = 11.5%

	2010	2009
Total assets	5,190	4,670
Cash and investments	(150)	(100)
Operating assets	5,040	4,570
Total liabilities	3,250	2,990
Notes payable	(200)	(220)
Long term debt	(770)	(680)
Operating liabilities	2,280	2,090
Net operating assets	2,760	2,480

(Module 13.5, LOS 13.d)

Question #126 - 127 of 140

Question ID: 1473183

Using the leading P/E ratio of 4.29, determine whether Retro Inc. is *most likely* under/overvalued based on its PEG ratio:

A) Overvalued because its PEG ratio is 3.46.



B) Undervalued because its PEG ratio is 0.29.

**Explanation**

The firm's PEG is $4.29 / 14.85 = 0.29$. Given the comparable group median, PEG of 0.82, it appears that Retro Inc. may be undervalued.

(Module 22.4, LOS 22.j)

Question #127 - 127 of 140

Question ID: 1473184

Regarding the Cery's comments on enterprise value multiples which are *most likely* correct:

<u>Comment 1</u>	<u>Comment 2</u>	
A) Correct	Correct	
B) Correct	Incorrect	
C) Incorrect	Incorrect	

Explanation

Since FCFF captures the amount of capital expenditures, it is more strongly linked with valuation theory than EBITDA. EBITDA will be an adequate measure if capital expenses equal depreciation expenses and working capital remains relatively constant.

(Module 21.5, LOS 21.h)

Question #128 of 140

Question ID: 1473155

What is the appropriate leading price-to-earnings (P/E) multiple of a stock that has a projected payout ratio of 40% if shareholders require a return of 15% on their investment and the expected growth rate in dividends is 5%?

- | | |
|-----------|---|
| A) 6.30. |  |
| B) 13.20. |  |
| C) 4.00. |  |

Explanation

$$\text{Justified leading P/E} = P_0/E_1 = (1 - b) / (r - g) = 0.40 / (0.15 - 0.05) = 4.00$$

Note that the leading P/E omits $(1 + g)$ in the numerator, which is present in the formula for the trailing P/E.

(Module 22.4, LOS 22.i)

Question #129 of 140

Question ID: 1473108

An analyst focusing mostly on financial stocks is likely to prefer valuing stocks via the:

- A) price/book ratio.
- B) price/sales ratio.
- C) dividend yield.



Explanation

The price/book ratio is a preferred tool for valuing financial stocks.

(Module 22.2, LOS 22.c)

Question #130 of 140

Question ID: 1473134

A method commonly used to normalize earnings is the method of:

- A) average return on assets.
- B) historical average earnings per share (EPS).
- C) comparables.



Explanation

A common method in normalizing earnings uses the historical average EPS.

(Module 22.4, LOS 22.e)

Question #131 of 140

Question ID: 1473095

The value of a firm, calculated using the discounted cash flow (DCF) method, will be closest to the valuation using P/E multiples when P/E multiples are estimated using:

- A) fundamental data.
- B) P/E multiples of comparable firms.
- C) historical P/E multiples.



Explanation

In the DCF valuation method, an analyst makes specific assumptions about each variable, such as growth, risk, payout, etc. The valuation using P/E multiples will be closest to the one obtained using the DCF approach when fundamental data -- for growth, risk, payout, etc. -- is used to estimate P/E multiples.

(Module 22.1, LOS 22.a)

Question #132 of 140

Question ID: 1473091

An analyst begins an equity analysis of Company A by estimating future cash flows, discounting them back to the present, and dividing the result by the outstanding number of shares. This analyst is *most likely* using the:

- A) the method of forecasted fundamentals.
- B) the method of comparables.
- C) technical analysis.



Explanation

This analysis is comparing forecasted discounted cash flows (DCF) to a fundamental variable (shares). This suggests the method for forecasted fundamentals.

(Module 22.1, LOS 22.a)

Question #133 of 140

Question ID: 1473103

An argument for using the price-to-earnings (P/E) valuation approach is that:

- A) research shows that P/E differences are significantly related to long-run average stock returns.
- B) earnings volatility facilitates interpretation.
- C) earnings can be negative.



Explanation

Research shows that P/E differences are significantly related to long-run average stock returns. Both remaining factors reduce the usefulness of the P/E approach.

(Module 22.1, LOS 22.c)

Question #134 of 140

Question ID: 1473156

An analyst has gathered the following data about Jackson, Inc.:

- Payout ratio = 60%.
- Expected growth rate in dividends = 6.7%.
- Required rate of return = 12.5%.

What will be the appropriate price-to-book value (PBV) ratio for Jackson, based on fundamentals?

- A) 0.58.
- B) 1.38.
- C) 1.73.



Explanation

Return on equity (ROE) = $g / (1 - \text{payout ratio}) = 0.067 / 0.40 = 0.1675$ or 16.75%.

Based on fundamentals:

PBV = $(0.1675 - 0.067) / (0.125 - 0.067) = 1.73$.

(Module 22.4, LOS 22.i)

Question #135 of 140

Question ID: 1586178

Herb McClain tells Cammy Oren that Kline Industries' expected dividend growth rate is 4.0%, ROE is 14%, and required return on equity (r) is 10%. Based on a justified P/B ratio compared to a P/B ratio (based on market price per share) of 1.55, Kline Industries is *most likely*:

A) correctly valued.



B) undervalued.



C) overvalued.



Explanation

Justified P/B = $(\text{ROE} - g) / (r - g)$. When the expected dividend growth is 4.0%, the justified P/B = $(0.14 - 0.04) / (0.10 - 0.04) = 1.67$. This is greater than the P/B (at market) of 1.55.

(Module 22.2, LOS 22.I)

Question #136 of 140

Question ID: 1473186

Precision Tools is expected to have earnings per share (EPS) of \$5.00 per share in five years, a dividend per share of \$2.00, a cost of equity of 12%, and a long-term expected growth rate of 5%. What is the terminal trailing price-to-earnings (P/E) ratio in five years?

A) 6.00.



B) 7.14.



C) 9.00.



Explanation

$P_5/E_5 = (0.40 \times 1.05) / (0.12 - 0.05) = 6.00$

(Module 22.4, LOS 22.k)

Question #137 of 140

Question ID: 1473179

Consider the statement: "Unlike many valuation metrics that incorporate dividend discounting, the PEG ratio may be used to value firms with zero expected dividend growth prospects." Is this statement correct?

A) No, because the PEG ratio is undefined for zero-growth companies.



B) Yes, because the computation of the PEG ratio does not use the rate of expected dividend growth.



C) Yes, because the expected dividend growth rate is cancelled out in the computation of the PEG ratio.



Explanation

The PEG ratio measures the tradeoff between P/E and expected earnings growth (g). The formula for the PEG ratio is: $PEG = (P/E) / g$. Firms with zero expected earnings growth will have an infinite (or undefined) PEG ratio due to division by zero.

(Module 22.4, LOS 22.j)

Question #138 of 140

Question ID: 1473152

A firm has a return on equity (ROE) of 18%, an estimated growth rate of 13%, and its shareholders require a return of 17% on their investment. Based on these fundamentals, a reasonable estimate of the appropriate price-to-book value ratio for the firm is:

A) 1.25.



B) 2.42.



C) 1.58.



Explanation

$$\frac{P}{BV} = \frac{ROE - g}{r - g} = \frac{0.18 - 0.13}{0.17 - 0.13} = 1.25$$

(Module 22.4, LOS 22.i)

Question #139 of 140

Question ID: 1473119

Bill Whelan and Chad Delft are arguing about the relative merits of valuation metrics.

Whelan: "My ratio is less volatile than most, and it works particularly well when I look at stocks in cyclical industries."

Delft: "The problem with your ratio is that it doesn't reflect differences in the cost structures of companies in different industries. I like to use a metric that strips out all the fluff that distorts true company performance."

Whelan: "People can't even agree how to calculate your ratio."

Which valuation metric do the analysts *most likely* prefer?

Whelan

Delft

A) Price/book EV/EBITDA



B) Price/cash flow Price/book



C) Price/sales Price/cash flow



Explanation

The price/sales ratio is not very volatile, and it is of particular value when dealing with cyclical companies. The price/cash flow ratio considers the stock price relative to cash flows, ignoring the noncash gains and losses that can skew earnings. A major weakness of the price/cash flow ratio is the fact that there are different ways of calculating it, making comparisons difficult at times.

(Module 22.3, LOS 22.c)

Question #140 of 140

Question ID: 1473214

In interpreting the standardized unexpected earnings (SUE) momentum measure, it can be concluded that a given size forecast error is:

A) more meaningful the larger the historical size of forecast errors.



B) scaled by the earnings surprise.



C) more meaningful the smaller the historical size of forecast errors.



Explanation

A given size forecast error is more (less) meaningful the smaller (larger) the historical size of forecast errors.

(Module 22.4, LOS 22.q)